Geophysical Journal International

Geophys. J. Int. (2011) **186,** 471–491 doi: 10.1111/j.1365-246X.2011.05030.x

Inclination variation in the Late Jurassic to Eocene red beds from southeast Asia: lithological to locality scale approach

Shun Sato,[1] Zhenyu Yang,[2] Yobo Tong,[3] Makoto Fujihara,[1] Haider Zaman,[4] Masahiko Yokoyama,[1] Kazuya Kitada[5] and Yo-ichiro Otofuji[1]

[1] *Department of Earth and Planetary Sciences, Faculty of Science, Kobe University, Kobe, Japan. E-mail: otofuji@kobe-u.ac.jp*
[2] *Department of Earth Sciences, Nanjing University, Nanjing, China*
[3] *Institute of Geomechanics, Chinese Academy of Geological Science, Beijing, China*
[4] *Department of Geology and Geophysics, College of Science, King Saud University, Riyadh, Saudi Arabia*
[5] *Institute for Research on Earth Evolution, Japan Agency for Marine-Earth Science and Technology, Yokosuka, Japan*

Accepted 2011 March 30. Received 2011 February 15; in original form 2010 August 24

SUMMARY

Shallow palaeomagnetic inclinations have been frequently reported from the red beds of central and southeast Asia. To trace the origin of this phenomenon, inclination variation in southeast Asia is examined on lithological to locality scale. Lithological aspect of this variation has been studied using the Early Cretaceous Bailong and Cangxi formations of the Bazhong area (32.1°N, 106.7°E), the northern Sichuan Basin. Samples from 36 sites, consisting 18 pairs of successive sandstones and mudstones layers, were collected for this purpose from a synclinal structure. Stepwise thermal demagnetization of most samples revealed the presence of stable characteristic remanent magnetization, which is generally unblocked by 680 °C. Positive fold and reversal tests suggest a primary origin for this component, yielding the Early Cretaceous palaeomagnetic direction of declination/inclination = 20.9°/26.5° (ks = 37.2, $\alpha_{95} = 4.4°$, $N = 30$). 10 pairs of sandstone and mudstone layers show almost identical inclinations ($I_{\text{sandstones}} = 23.3° \pm 3.7°$ and $I_{\text{mudstones}} = 24.7° \pm 2.4°$), but 27° shallower than that expected from the Eurasian apparent polar wander paths (APWPs), indicating that no lithological variation in inclination has occurred. Location-wise variation in inclination shallowing is examined through palaeomagnetic data from Late Jurassic to Eocene red beds distributed around southeast Asia. Based on these investigations, no inclination shallowing is observed in the eastern part of the South China Block (SCB), whereas large degree of shallowing is observed in the Sichuan and Xining-Ninhe basins. Variation in inclination shallowing from one sedimentary basin to another could probably be caused by changes in the depositional environment. The eastern part of the SCB, where no inclination shallowing have been observed, is characterized by Basin and Range type tectono-geological setting. In contrast, the foreland basins are bounded to the north by east–west striking high altitude orogenic belt. Swift development of these basins as a result of flexural subsidence is the most likely reason for inclination shallowing.

Key words: Palaeomagnetism applied to tectonics; Palaeomagnetism applied to geologic processes; Rock and mineral magnetism.

1 INTRODUCTION

After the initial appearance of low palaeolatitudes in the Mesozoic to Lower Tertiary rocks of the Tethyan belt in Asian Continent (Westphal *et al*. 1986), shallow inclinations have been frequently reported from the red beds of North America, the Alpine–Himalaya mountain belt and the central–east Asian blocks (e.g. Gilder *et al*. 2003; Tauxe 2005; Narumoto *et al*. 2006; Tan *et al*. 2007; Wang & Yang 2007; Tauxe *et al*. 2008). Palaeomagnetic results from these red beds reveal inclination shallowing between 10° and 30° compared to those expected from the reference APWPs. Because of globally distributed inclination shallowing, the nature of geomagnetic field has been advocated as a possible cause; particularly the role of axial octupole (g_0^3) in the geocentric dipole has been stressed in this regard (Westphal 1993; Van der Voo & Torsvik 2001). Alternatively, as a common phenomena in the red beds, inclination shallowing has been ascribed to a bias in recording during the process of sedimentation and

© 2011 The Authors
Geophysical Journal International © 2011 RAS

GJI Geomagnetism, rock magnetism and palaeomagnetism
Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

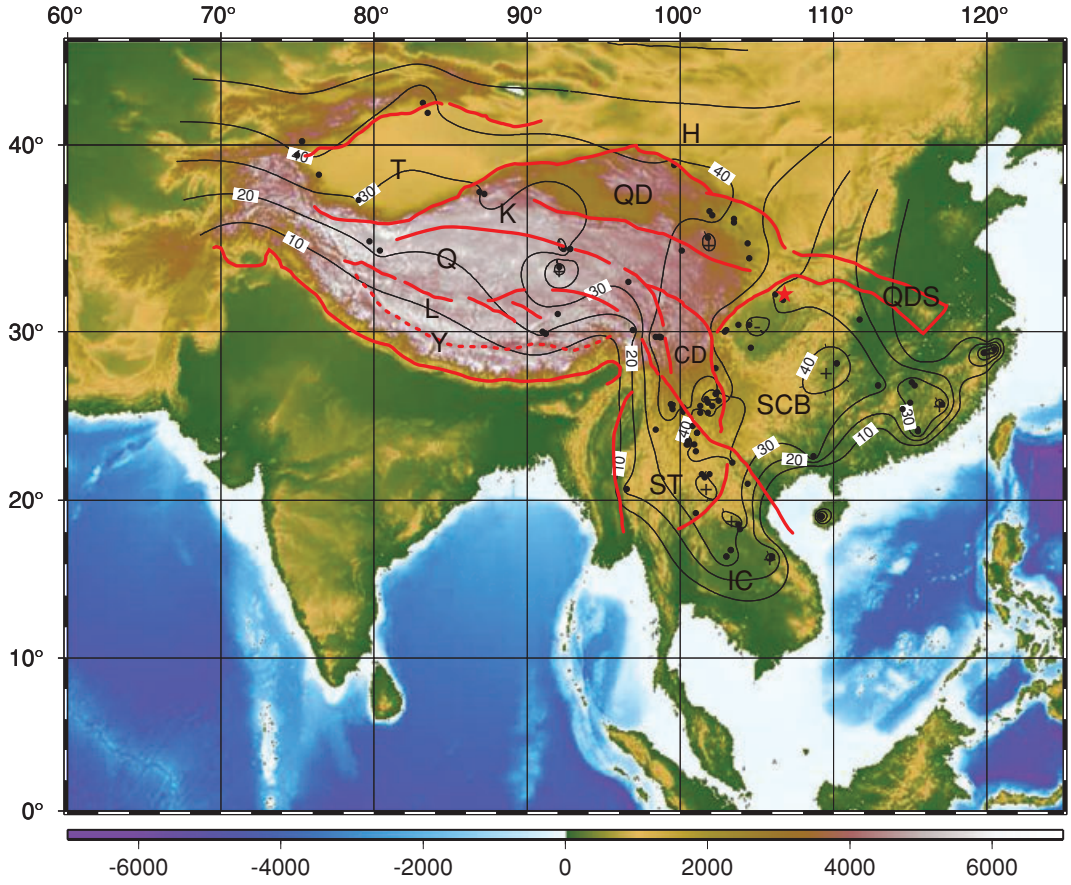


Figure 1. Contours map of palaeomagnetic inclinations together with topographic map of eastern Himalayan syntaxis and surrounding areas. Tectonic zonations are modified from Tapponnier *et al*. (2001): CD, Chuan Dian Fragment; H, Hexi Corridor; IC, Indochina Block; K, Kunlun Block; L, Lhasa Block; Q; Qiangtang Block; QD, Qaidam Block; QDS, Qinlin-Dabie Suture Zone; SCB, South China Block; ST, Shan-Thai Block; T, Tarim Block; Y, Yarlung Zangbo Suture Zone. Closed circles represent palaeomagnetically studied areas of the Jurassic to Eocene rocks as listed in Table 2. The present study area is shown by a red star. The bathymetry and land topography are derived from ETOPO2 global data set (National Geophysical Data Center 2001).

subsequent compaction (Tauxe & Kent 2004; Tan *et al*. 2007). This is in sharp contrast to TRM acquisition in the basaltic rocks and its comparison to reference directions of the APWPs (e.g. Bazhenov & Mikolaichuk 2002; Hankard *et al*. 2007a).

Several palaeomagnetists have attempted to document regional inclination variation in the Cretaceous red beds of central Asia, where systematic inclination shallowing (10–20°) have been observed as compared to the reference APWPs. Several possible causes have been suggested in this connection, that is, local magnetic field anomaly at Eurasian scale (Chauvin *et al*. 1996), tectonic shortening in the Asian continent (Chen *et al*. 1993; Hankard *et al*. 2007b) and non-rigid nature of the Eurasian Plate (Cogné *et al*. 1999).

As shown in Fig. 1, locality-based variation in inclination shallowing has been observed in the Cretaceous red beds of the South China Block (SCB). Although, inclination shallowing has not been detected in the coastal area of the SCB (Morinaga & Liu 2004; Zhu *et al*. 2006), anomalously large shallowing has been reported from the Sichuan Basin of the same block (Otofuji *et al*. 1990; Enkin *et al*. 1991a). Following these observations, demand for further investigations became high to trace local origin of the inclination shallowing.

Small-scale variation of inclination shallowing can be observed in the red beds strata, which mainly depends on the distribution of the size of grains and their flocculated matters (flocs) in the hematite bearing sediments as well as their mechanical compaction (Payen & Verosub 1982; Tan *et al*. 2002; Heslop 2007; Mitra & Tauxe 2009). Accordingly, lithological variation on strata scale could be taken as one of the targets in studying inclination shallowing.

Palaeomagnetic study of the red beds on lithological to locality scale can give us a clue to investigate the origin of inclination shallowing. As a first step in this procedure, palaeomagnetic study have been conducted to detect grain size dependence of inclination shallowing. As a second step, locality-wise features of inclination shallowing in the southeast Asia are investigated. For this purpose, the Early Cretaceous red beds of the Sichuan Basin have been targeted, where large degree of inclination shallowing has been observed.

2 GEOLOGICAL SETTING AND SAMPLING

The Sichuan Basin, which is filled by the Mesozoic continental red beds, is located in the northwestern part of the SCB (Figs 1 and 2). According to a report from the Bureau of Geology and Mineral Resources of Sichuan Province (1991, hereafter BGMRSP), red beds of the Early Cretaceous age are distributed in the northwestern part of this basin. These rocks belong to Chengqiangyan Group, which consists of fluviolacustrine facies

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

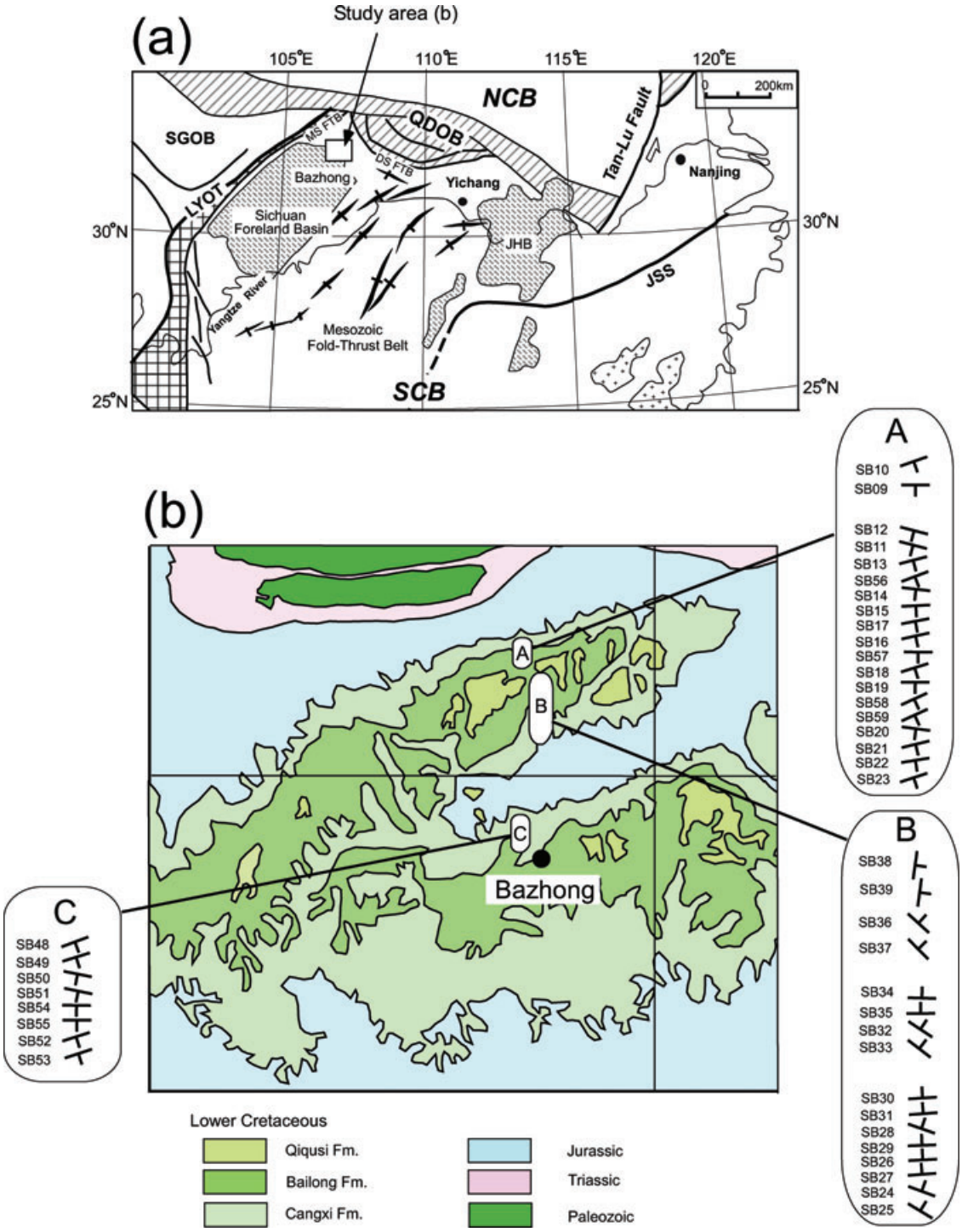


Figure 2. (a) Simplified tectono-geological map of eastern China (after Su *et al*. 2005), and (b) geological map of the study area (after BGMRSP). NCB, North China Block; SCB, South China Block; QDOB, Qinling-Dabie Orogenic Belt; MS FTB, Micang Shan fold-thrust belt; DS FTB, Daba Shan fold-thrust belts; SGOB, Songpan-Guanze fold belt; LYOT, Longmenshan-Yanyuan over thrust fault; JSS; Jingshan-Shaoxing Suture. (b) Three sampling localities (A, B and C) are shown by boxes. Strike and dip of strata at each sampling site is shown in the insets.

deposits of the Cangxi, Bailong, Qiqusi and Gudian formations in ascending order. The rocks of Cangxi and Bailong formations cropped out near the Bazhong city (Fig. 2). The lower part of the Early Cretaceous Cangxi Formation, which disconformably overlies the Jurassic formation, consists of greyish-purple (or green-purplish grey) and fine-to-coarse grained sandstone intercalated with thin layers of reddish to brown siltstones and mudstones. The overlying Bailong Formation of the Early Cretaceous age consists of purple to yellowish grey fine-to-coarse grained sandstones and purple to brownish siltstones and mudstones.

The Upper Jurassic and Lower Cretaceous red beds of this basin are gently folded. The age of folding in this area has been placed

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

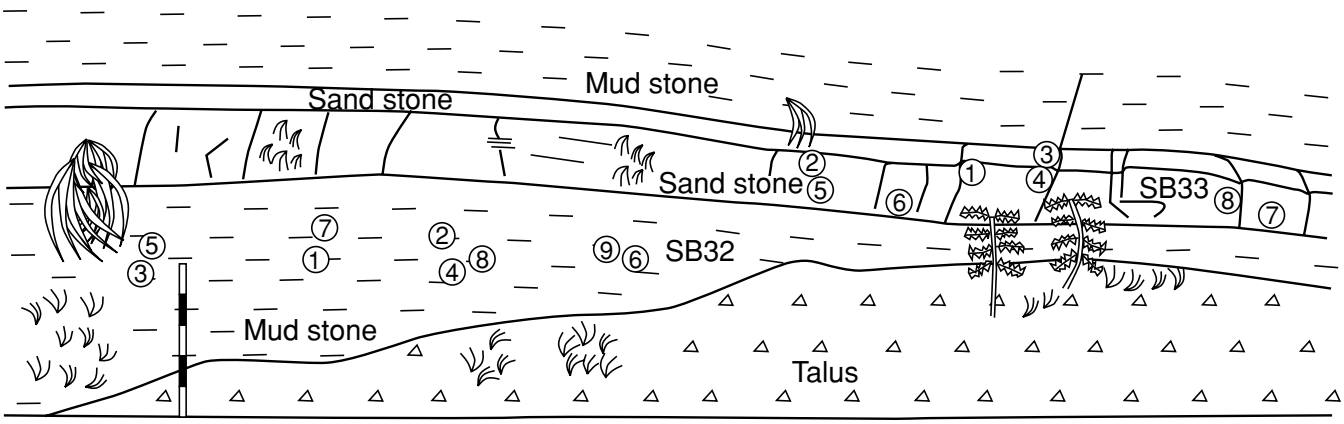


Figure 3. Sketch map of the selected pair of sampling sites from mudstone (SB32) and sandstone (SB33) with a scale (1 m). Numbers on the map correspond to hand samples.

at Palaeogene, because the Early Neogene rocks unconformably overlies the Jurassic to Palaeogene strata.

Palaeomagnetic samples were collected in 2005 from the Early Cretaceous Bailong and Cangxi formations at three different localities (A, B and C) near the city of Bazhong (32.1°N, 106.7°E), where two synclinal structures with east–west trending axes have been observed (Fig. 2). Sampling areas A and B are located on the northern and southern limbs of one syncline, respectively, while area C is situated on the northern limb of another syncline. Palaeomagnetic samples were collected at 40 sites from A (16 sites), B (16 sites) and C (8 sites) localities. In 36 of 40 sites, 18 pairs of successive sandstones and mudstones layers have been observed (Fig. 3). Location of each site was determined by portable navigational system, whereas samples were oriented by magnetic compass. The present declination value at each sampling site was evaluated from the International Geomagnetic Reference Field (Macmillan & Maus 2005).

3 LABORATORY PROCEDURES

In the palaeomagnetic laboratory of the Kobe University, individual specimens with 25 mm diameter and 23 mm length were prepared from the collected samples. Because of their fragile nature, only small number of specimens could be prepared from several mudstones layers, while no specimen could be obtained from site SB14.

Natural remanent magnetizations (NRMs) were measured with 2-G Enterprises cryogenic magnetometer. At least one specimen from each sample was subjected to stepwise thermal demagnetization procedure up to 690 °C using Natsuhara TDS-1 thermal demagnetizer (with a residual field of less than 5 nT). Results from demagnetization procedure were plotted on orthogonal vector diagrams (Zijderveld 1967) to assess component structure as well as on equal-area projections to evaluate directional stability. Principal component analysis (Kirschvink 1980) was used to estimate the component directions, and site-mean direction was calculated using Fisherian statistics (Fisher 1953).

Progressive acquisition of isothermal remanent magnetization (IRM) was performed up to a maximum field of 2.7 T using 2G-pulse magnetizer. Specimens with acquired IRM were then subjected to back field demagnetization. To determine the unblocking temperature spectra, thermal demagnetization of the composite IRMs (2.7, 0.4 and 0.12T along z, y and x axes of specimens, respectively) were carried out (Lowrie 1990). Anisotropy of magnetic susceptibility (AMS) measurements were made on sandstones and mudstones specimens using Kappabridge KLY-3 apparatus, and then analysed by the method of Tarling & Hrouda (1993). Principal susceptibilities (maximum $= K_1$, intermediate $= K_2$ and minimum $= K_3$) were calculated for each specimen. Features of the magnetic fabric were determined by shape parameter (T) and a degree of anisotropy (Pj).

4 ROCK MAGNETIC RESULTS

As evident from IRM acquisition curves (Fig. 4), no saturation is achieved up to a maximum field of 2.7 T, indicating the presence of high-coercivity magnetic minerals, such as hematite or goethite. The presence of hematite is clearly confirmed by unblocking temperature spectra (between 660 and 690 °C) of the hard component obtained by thermal demagnetization of the composite IRMs (Fig. 4).

Samples from mudstone and sandstone are clearly discriminated by IRM acquisition and back field demagnetization experiments. As demonstrated by logarithmic IRM/SIRM ratio (Fig. 4), mudstone samples are magnetically harder compared to sandstone samples. This may be ascribable to grain size effect on the coercivity spectra (Thompson 1986), as mudstones usually carry finer hematite grains. This implication is also supported by stepwise thermal demagnetization of the composite IRMs, where samples from mudstone reveal broader unblocking temperature spectra compared to sandstone (Fig. 4). These features can fairly be compared with the distributed and discrete unblocking temperature spectra reported by Tan *et al*. (2007) for hematite bearing mudstones and sandstones. The studied features also resemble to demagnetization curves established by Dekkers & Linssen (1989) for hematite bearing rocks with different grain size fractions.

The AMS results from the studied samples predominantly shows an oblate ellipsoid (Fig. 5). The tilt-corrected AMS data indicate the minimum axis nearly perpendicular to a bedding plane, while the maximum and intermediate axes are dispersed on a bedding plane. This type of distribution reveals the presence of sedimentary origin fabric in the studied rocks, with no significant deformational effects.

As shown in the Fig. 5, all the studied specimens from mudstones but one show low degree of AMS (sandstones; $1.041 < Pj < 1.140$, and mudstones; $1.028 < Pj < 1.137$), which is comparable to those observed in the Shaxian and Chongan formations of the Southeast China Fold Belt ($1.008 < Pj < 1.107$), where no inclination shallowing have been observed (Morinaga & Liu 2004; Narumoto *et al*. 2006).

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

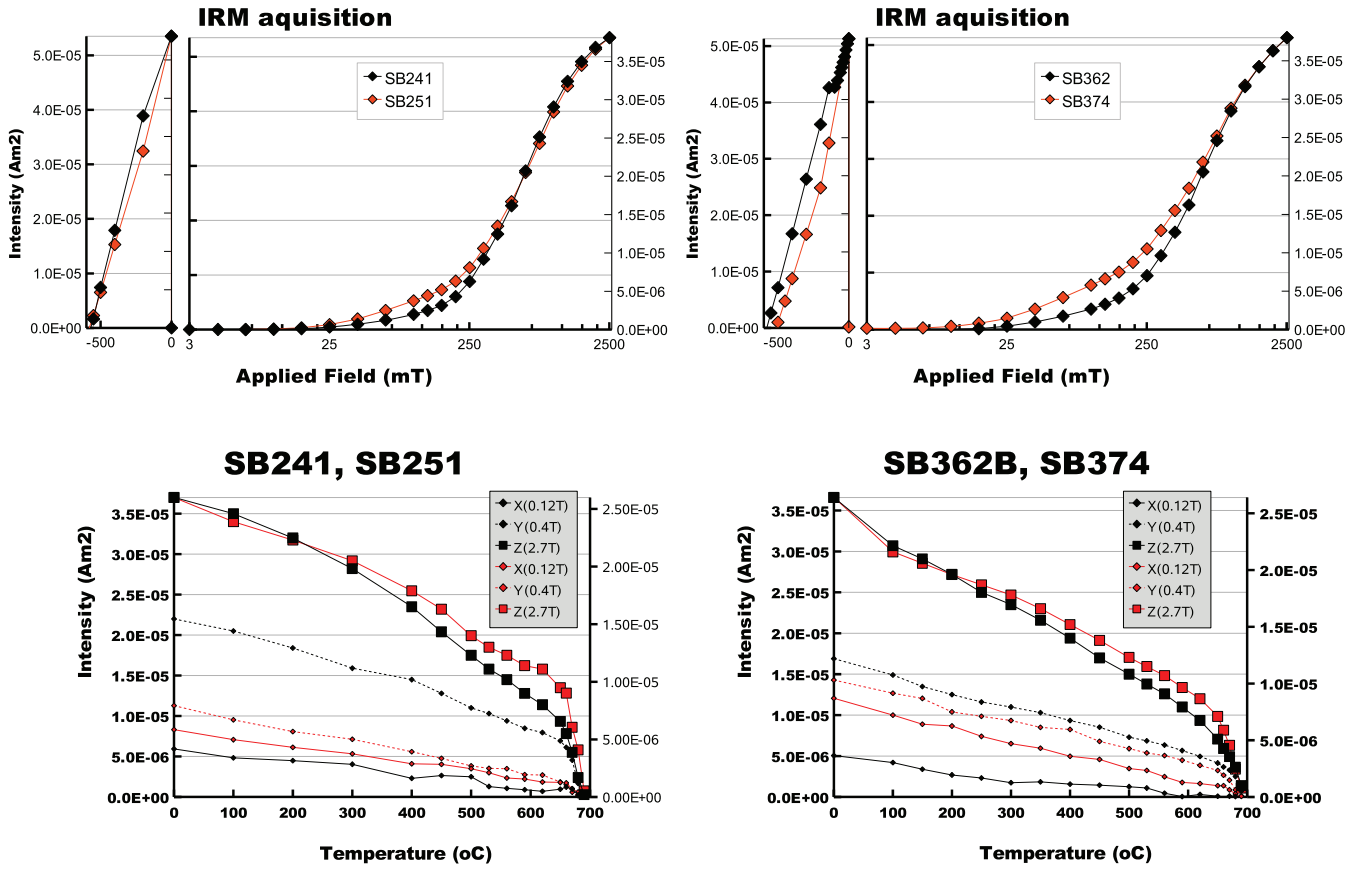


Figure 4. IRM acquisition and back-field demagnetization curves are shown in the left box. Thermal demagnetization of the composite IRMs (2.7, 0.4 and 0.12 T in DC field along three perpendicular axes of the sample) for representative paired sites (SB24-SB25 and SB36-SB37) are shown in the right box. The studied specimens SB241, SB251, SB362 and SB374 are selected from sites SB24, SB25, SB36 and SB37, respectively. All specimens received gradual increase in IRM intensity up to 2.7 T DC field, but could not be completely saturated. Thermal demagnetization of three components IRM show an unblocking temperature of hematite. The red colour mudstone show magnetically hard component with distributed spectra of unblocking temperature, while black colour sandstones indicate the presence of magnetically soft component with discrete spectra.

Although, the origin of anomalously high degree of AMS ($Pj = 1.532$) in one of the mudstone specimen (SB327) remained unclear, the presence of disseminated poyllsilicate in the rock matrix might be the reason rather than depositional and/or compaction related fabrics (Evans *et al*. 2003).

5 PALAEOMAGNETIC RESULTS

To revisit different aspects of inclination shallowing in the red beds of central and east Asia, NRMs of 224 samples from 40 sites have been measured for this study. These samples have been exclusively collected from Bailong and Cangxi formations, of which 22 sites are assigned to sandstones 18 to mudstones.

5.1 Variation in the NRM intensities

Initial NRM intensities in the studied samples are in a range between 0.6×10^{-3} and 3.7×10^{-2} A m^{-1}, where sandstone samples gave slightly lower intensity than that of the mudstone samples, that is, $7.8 \pm 1.3 \times 10^{-3}$ and $8.2 \pm 1.5 \times 10^{-3}$ A m^{-1}, respectively. About half of the sandstone samples carry intensities of less than 5.0×10^{-3} A m^{-1}, while only one third of mudstone's samples reveal a similar trend. The *F*-test applied to studied samples, however, show almost similar variance in distribution of the NRM intensity between the sandstone and mudstone samples at 95 per cent confidence limit. Comparison of the mean NRM intensities between 17 pairs of sandstone and mudstone layers indicate statistically identical behaviour at 95 per cent confidence limit, such as, $8.1 \pm 4.2 \times 10^{-3}$ and $8.5 \pm 2.9 \times 10^{-3}$ A m^{-1}, respectively.

5.2 Demagnetization behaviour

Almost single component behaviour has been observed in the samples with initial NRM intensity larger than 1.1×10^{-2} A m^{-1}, which was unblocked in a temperature range between 680 and 690 °C (Fig. 6). As the NRM intensity decreases, the two components behaviour appeared. After the isolation of low-temperature component (LTC) by 350 °C, the high-temperature component (HTC) appeared and was then unblocked between 650 and 690 °C. An unclear HTC is observed in low intensity samples as well. In samples with less than 1.5×10^{-3} A m^{-1} NRM intensity (e.g. SB17, SB19 and SB52), only spurious demagnetization behaviour has been observed.

5.3 Directional behaviour of the NRM

LTC of the NRM is obtained from 117 samples, including 66 from sandstone and 49 from mudstone. Mean directions calculated for this component in geographic coordinates are $D = 355.0°$, $I = 47.1°$ ($k = 10.7$, $\alpha_{95} = 5.5°$, $N = 68$) for sandstone samples and $D = 356.4°$, $I = 47.4°$ ($k = 8.7$, $\alpha_{95} = 7.3°$, $N = 49$) for mudstone

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

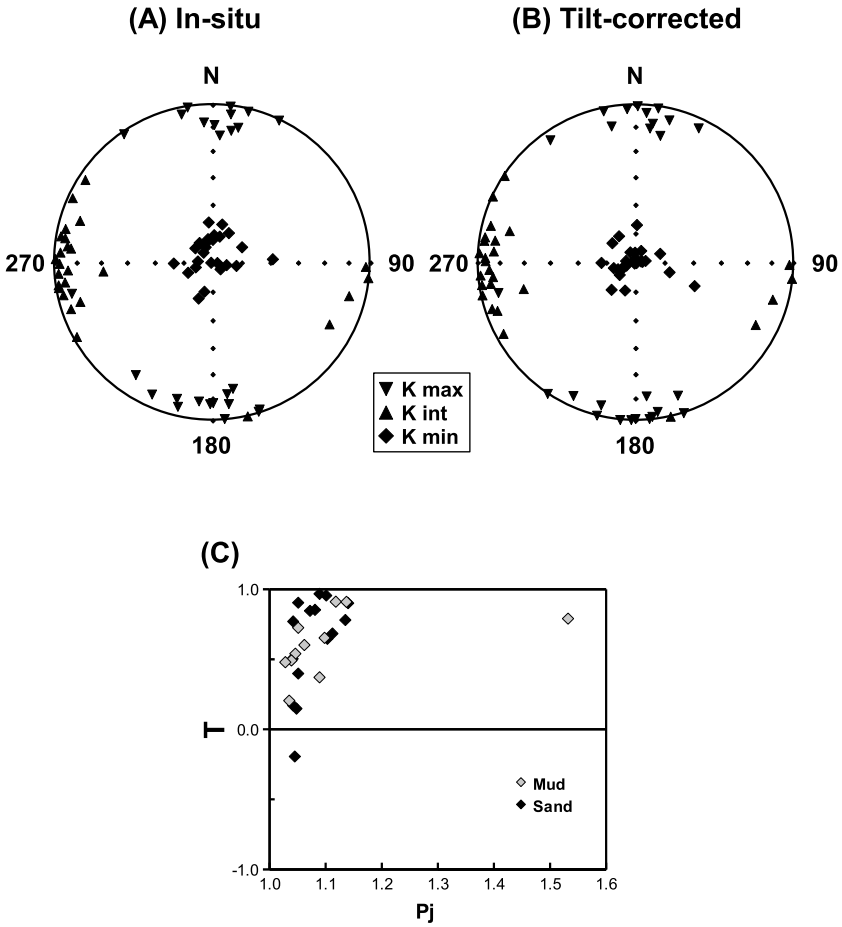


Figure 5. (A and B) Principal directions of magnetic susceptibility. (C) Plots of shape parameter (T) and degree of anisotropy (Pj). Principal directions are projected onto lower hemisphere through equal-area projection. Squares are maximum susceptibility direction (K_1); triangles are intermediate susceptibility directions (K_2); circles are minimum susceptibility directions (K_3).

samples, which look almost identical to present geocentric axial dipole moment ($D = 0^\circ$, $I = 51.4^\circ$) and the IGRF direction ($D = 357.7^\circ$, $I = 48.9^\circ$) in the study area. A direction-correction (DC) tilt test (Enkin 2003) gives an optimal concentration at -6.5 ± 37.8 per cent unfolding for sandstones and at 5.8 ± 69.8 per cent unfolding for mudstone samples, which are indistinguishable from 0 per cent unfolding. The LTC in the studied samples is, therefore, considered as secondary magnetization acquired during the Bruhnes Chron.

The HTC is identified in 31 of 40 sites (Table 1), but data from one site (SB31) was discarded for further analysis due to large dispersion ($k = 8.5$) and unexpected deflection in declination with respect to north. 10 of 30 sites gave normal polarity directions and the remaining 20 sites revealed reversed polarity directions. After flipping the reversed polarity directions into normal, formation mean direction of 30 sites is calculated as declination/inclination (dec./inc.) = $20.8^\circ/19.3^\circ$ ($kg = 36.4$, $\alpha_{95} = 4.4^\circ$) in geographic coordinates and dec./inc. = $20.9^\circ/26.5^\circ$ ($ks = 37.2$, $\alpha_{95} = 4.4^\circ$) in stratigraphic coordinates (Fig. 7).

Reliable palaeomagnetic directions have been extracted from 17 sandstones layers (sites) and 13 mudstone layers (Fig. 8), which gave tilt-corrected formation mean directions of dec./inc. = $21.9^\circ/27.7^\circ$ ($ks = 35.5$, $\alpha_{95} = 6.1^\circ$, N = 17) and dec./inc. = $19.6^\circ/25.0^\circ$ ($ks = 37.9$, $\alpha_{95} = 6.8^\circ$, $N = 13$), respectively. A difference in palaeomagnetic directions between the sandstone and mudstone layers (sites) is only $4.0^\circ \pm 8.8^\circ$, indicating almost identical directional behaviour within 95 per cent confidence limit.

Variation of inclinations with respect to polarity is also examined. The sandstone layers with reversed polarity directions shows shallower inclinations than that with normal polarity directions ($I_{\text{normal}} = 31.5^\circ$, $\alpha_{95} = 6.8^\circ$, $N = 5$ versus $I_{\text{reversed}} = 26.0^\circ$, $\alpha_{95} = 7.9^\circ$, $N = 12$), whereas an opposite relationship is observed in the mudstones layers ($I_{\text{normal}} = 21.5^\circ$, $\alpha_{95} = 13.7^\circ$, $N = 5$ versus $I_{\text{reversed}} = 27.1^\circ, \alpha_{95} = 8.9^\circ, N = 8$). Comparison between the normal and reversed polarity directions indicate no significant inclination shallowing in the reversed polarity sites.

Due to identical directional behaviour of the HTC in sandstone and mudstone layers, fold and reversal tests were applied to all reliable sites from these layers. Sites from localities A and B are particularly chosen for fold test, because they stand on northern and southern limbs of the same syncline, respectively. Mean direction calculated on the basis of 23 sites from this syncline is $D = 21.5^\circ$, $I = 18.9^\circ$ ($kg = 40.5, \alpha_{95} = 4.8^\circ$) in geographic coordinates and $D = 21.1^\circ, I = 24.4^\circ$ ($ks = 47.4, \alpha_{95} = 4.4^\circ$) in stratigraphic coordinates. The DC tilt-test of Enkin (2003) gives an optimal concentration at 65.2 ± 44.2 per cent unfolding, which is indistinguishable from that at 100 per cent unfolding. Following the fold tests of McFadden's (1990) and Cogné (2003), a value (ξ_1) of 8.011 is obtained in the geographic coordinates and 2.437 in the stratigraphic coordinates, while critical values (ξ_c) of 5.583 and 7.836 are calculated at 95 and 99 per cent confidence levels, respectively.

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

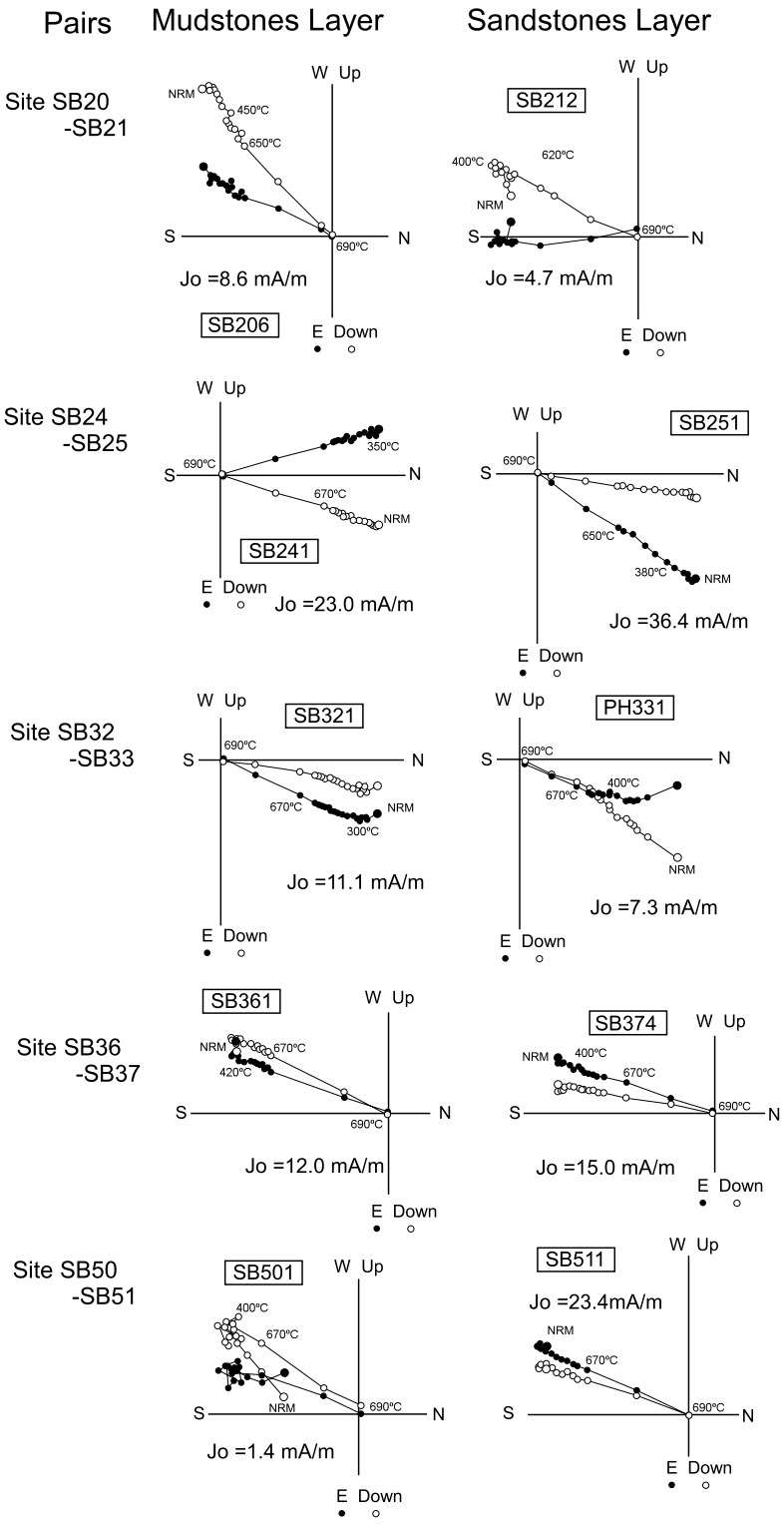


Figure 6. *In situ* vector end-point diagrams of representative specimens from the paired sites of red colour mudstone and sandstone (SB20-SB21, SB24-SB25, SB32-SB33, SB36-SB37 and SB50-SB51) after stepwise thermal demagnetization. Solid (open) symbols are projections on to horizontal (vertical) plane.

On the basis of the data from all three localities, the tilt-corrected normal (dec./inc. = 22.1°/26.6°, $kg = 37.4$, $\alpha_{95} = 8.0°$, $N = 10$) and reversed (dec./inc. = 200.3°/–26.5°, $ks = 35.5$, $\alpha_{95} = 5.6°$, $N = 20$) polarity mean directions are estimated, revealing a passage of reversal test (McFadden & McElhinny 1990) with classification 'A'. An angular distance (γ) obtained by this procedure is 1.6°, which is less than a critical angle of $\gamma_c = 9.4°$. A positive reversal test at 95 per cent confidence level implies that the effects of geomagnetic secular variation have been sufficiently averaged out. All these observations, thus, suggest that remanent magnetization carried by HTC in the Bailong and Cangxi formations is pre-folding and primary in origin.

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

Table 1. Palaeomagnetic results of the red bed samples collected at three localities (A, B and C) from the Lower Cretaceous Bailong and Cangxi formations, around the Bazhong City (32.1°N, 106.7°E). Magnetization directions for high-temperature component are listed in this table. Lat. and Long., correspond to latitude and longitude. SS and MS are sandstones and mudstones, respectively. 10 pairs of sites are denoted by I to X. *N* and *n* are number of samples measured and used for calculation, respectively. Dec. and Inc. are declination and inclination, respectively; *k* is the Fisherian precision parameter (Fisher 1953); α_{95} is the radius of cone at 95 per cent confidence level about the mean direction. The site with asterisk (SB31*) is excluded from mean direction calculation.

	Location									In situ		Tilt corrected			
Site	Lat. (°N)	Lon. (°E)	Strike (°)	Dip (°)	Formation lithology		Pair	*n*/*N*	Polarity	Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)	*k*	α_{95} (°)
Location A															
SB10	32.2229	106.7074	107.7	20.0	Cangxi	SS		6/6	N	17.8	26.4	17.8	46.4	41.4	10.5
SB12	32.2170	106.7109	109.7	16.0	Bailong	SS		6/6	R	196.7	–13.8	196.3	–29.7	58.6	8.8
SB56	32.2163	106.7116	104.7	16.0	Bailong	SS		6/6	R	198.6	–14.5	191.1	–30.4	73.6	7.9
SB14	32.2154	106.7122	98.7	14.0		MS		0/0							
SB15	32.2154	106.7122	92.7	13.8	Bailong	SS		6/6	R	210.9	–16.0	213.6	–28.0	36	11.3
SB16	32.2144	106.7129	101.7	12.5	Bailong	MS		5/5	R	210.7	–4.4	211.5	–16.2	107	7.4
SB17	32.2144	106.7128	101.7	12.5		SS		2/4							
SB57	32.2128	106.7153	92.7	15.5	Bailong	SS		4/6	R	192.2	–7.5	192.9	–22.8	19.4	21.4
SB18	32.2122	106.7162	105.7	11.0	Bailong	MS		1/3							
SB19	32.2121	106.7164	85.8	8.1	Bailong	SS		0/3							
SB58	32.2116	106.7167	114.7	12.0	Bailong	MS	I	6/6	R	206.9	–23.4	207.1	–35.3	75.3	7.8
SB59	32.2119	106.7168	114.7	12.0	Bailong	SS	I	5/6	R	195.2	0.5	195.0	–11.3	21.8	16.7
SB20	32.2113	106.7173	85.7	11.0	Bailong	MS	II	6/6	R	202.9	–23.4	205.7	–33.0	20.2	15.3
SB21	32.2109	106.7173	103.7	11.0	Bailong	SS	II	6/6	R	187.7	–16.7	187.2	–27.7	21.2	14.9
SB22	32.2106	106.7179	99.7	13.8	Bailong	MS	III	4/6	R	188.0	–10.8	187.9	–24.6	25.7	18.5
SB23	32.2104	106.7182	109.1	10.0	Bailong	SS	III	6/6	R	210.2	–8.2	210.6	–18.0	37.3	11.1
Location B															
SB38	32.1156	106.7527	172.2	2.5	Bailong	MS	VII	6/6	R	205.0	–18.2	204.3	–19.6	37.0	11.2
SB39	32.1156	106.7527	172.2	2.5	Bailong	SS	VII	6/6	R	198.4	–14.6	197.8	–15.6	135	5.8
SB36	32.1105	106.7594	227.7	3.0	Bailong	MS	VI	6/6	R	208.7	–24.7	207.4	–23.6	86.9	7.2
SB37	32.1105	106.7594	227.7	3.0	Bailong	SS	VI	6/6	R	201.4	–11.7	200.9	–10.4	381	3.4
SB34	32.0941	106.7687	269.2	5.3	Cangxi	MS		1/1							
SB35	32.0941	106.7687	269.2	5.3	Cangxi	SS		6/6	N	29.0	33.4	27.5	28.8	214	4.6
SB32	32.0913	106.7686	234.7	6.8	Cangxi	MS	V	6/6	N	23.0	15.0	21.6	11.4	54	9.2
SB33	32.0913	106.7686	234.7	6.8	Cangxi	SS	V	6/6	N	23.5	28.6	20.6	25.0	61.7	8.6
SB30	32.0813	106.7785	274.2	6.3	Cangxi	MS		7/7	N	38.5	45.3	35.3	40.0	14.5	25.0
SB31*	32.0813	106.7785	274.2	6.3	Cangxi	SS		6/6	N	91.6	48.6	84.6	47.9	8.5	24.5
SB28	32.0810	106.7783	248.7	4.5	Cangxi	MS		6/6	N	16.3	20.2	15.4	16.6	20.7	15.1
SB29	32.0809	106.7782	273.7	4.5	Cangxi	SS		3/3							
SB26	32.0795	106.7766	273.7	9.0	Cangxi	MS		3/3							
SB27	32.0794	106.7764	273.7	9.0	Cangxi	Silt		3/3							
SB24	32.0792	106.7761	238.7	10.0	Cangxi	MS	IV	6/6	N	9.8	27.9	6.9	20.2	28.2	12.8
SB25	32.0792	106.7761	249.2	10.5	Cangxi	SS	IV	7/7	N	28.7	28.2	25.1	21.2	14.7	18.0
Location C															
SB48	31.9198	106.7456	109.7	15.5	Cangxi	MS	VIII	6/7	N	21.7	3.0	21.8	18.5	12.0	23.1
SB49	31.9196	106.7457	107.7	15.5	Cangxi	SS	VIII	6/6	N	29.8	20.5	31.7	35.6	30.5	12.3
SB50	31.9188	106.7454	79.7	13.5	Bailong	MS	IX	6/6	R	193.1	–20.4	195.9	–32.7	29.8	12.5
SB51	31.9185	106.7456	75.7	11.0	Bailong	SS	IX	7/7	R	206	–11.2	208.0	–19.5	102	6.7
SB54	31.9172	106.7457	93.1	12.8	Bailong	MS	X	5/5	R	175.8	–15.5	175.1	–28.2	29.0	14.4
SB55	31.9172	106.7457	93.1	12.8	Bailong	SS	X	6/6	R	193.1	–36.0	195.3	–48.5	19.0	15.8
SB52	31.9147	106.7459	107.7	15.0	Bailong	MS		2/4							
SB53	31.9147	106.7459	107.7	15.0	Bailong	SS		6/6	R	210.4	–33.7	213.6	–48.3	60.2	8.7

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

Table 1. (*Continued.*)

Mean calculation	Location		In situ			Tilt corrected			
	Lat. (°N)	Lon. (°E)	n	Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)	k	α_{95} (°)
All data	32.1	106.8	30	20.8	19.3	10.9	26.5	36.4	4.4
			30					37.2	4.4
VGP	32.1	106.8	30			64.ON	234.8E		3.6
sandstone	32.1	106.8	17	21.7	19.0	21.9	27.7	41.4	5.6
								35.5	6.1
Mudstones	32.1	106.8	13	19.7	19.7	19.6	25	29.5	7.8
								37.9	6.8
Pairs of I – X Sandstone	32.1	106.8	10	21.5	17.7	21.3	23.4	38.2	7.9
								35.8	8.2
Mudstones	32.1	106.8	10	17.5	18.5	17.4	25.1	42.9 43.5	7.5 7.4

5.4 Data from 10 pairs of sandstone and mudstone layers

Since, 10 pairs of sandstone-mudstone layers provide reliable palaeomagnetic directions, their respective intensity (J) and inclination (I) values are compared in Fig. 9. Compared to sandstone layers, six mudstone layers from these pairs show higher NRM intensities, however, mean intensities of these layers are statistically identical, that is, $9.9 \pm 7.2 \times 10^{-3}$ A m^{-1} for sandstone and $9.3 \pm 4.1 \times 10^{-3}$ A m^{-1} for mudstone. A best-fitting line based on linear regression for 10 pairs is represented by $J_{\text{sandstones}} = -0.07 + 1.13 \times J_{\text{mudstones}}$, which gave a moderate correlation coefficient of 0.65. Concordance in the NRM intensities between sandstone and mudstone layers can be explained by similar source of inflow into the areas of deposition (river or lake).

Generally, no systematic difference in inclination has been observed between the sandstone and mudstone layers (Fig. 8). Although, six sandstones layers from 10 pairs gave steeper inclination than that of the mudstone layers, arithmetic means of inclinations for 10 sandstone–mudstone layers are almost identical at 95 per cent confidence limit; $I_{\text{sandstones}} = 23.3^\circ \pm 8.3^\circ$ and $I_{\text{mudtones}} = 24.7^\circ \pm 5.4^\circ$.

The observed inclinations in each pair of sandstone and mudstone layers show little concordance with others, where a correlation coefficient of only 0.09 is obtained. Inclination in the sandstone layers seems to be independent from the adjacent mudstone layers. However, taking in to consideration the 95 per cent confidence level, 9 of 10 pairs (except the pair SB36–37) show almost identical inclination behaviour.

6 DISCUSSION

Based on the positive fold and reversal tests, a primary palaeomagnetic direction is identified in the Early Cretaceous rocks of Bailong and Cangxi formations. As shown in Fig. 6, a tilt-corrected Early Cretaceous mean direction of dec./inc. = 20.9°/26.5° (ks = 37.2, $\alpha_{95} = 4.4^\circ$) is estimated by this study. This observed direction from Bazhong area (marked as Dobs, Iobs) is compared with 120Ma pole position expected ($D_{\text{ex}} = 14.0^\circ$, $I_{\text{ex}} = 52.3^\circ$) from the European APWPs of Besse & Courtillot (2002). Although, a difference in declinations appears less pronounce ($D_{\text{obs}} - D_{\text{ex}} = 6.9^\circ$), an observed mean inclination (26.5°) is significantly shallower than the expected inclination (52.3°) by 25.8° ± 4.0° (Flattening = $I_{\text{ex}} - I_{\text{obs}}$). It is, therefore, concluded that the Early Cretaceous palaeomagnetic direction from the Bazhong area is characterized by high degree of inclination shallowing.

6.1 Inclination shallowing in sandstone and mudstone layers

Although, inclination shallowing have been observed both in the sandstone and mudstone layers, their respective values are almost identical. Mean palaeomagnetic direction estimated for sandstones layers is dec./inc. = 21.9°/27.7°, $ks = 35.5$, $\alpha_{95} = 6.1^\circ$, $N = 17$ and that for mudstone layers is dec./inc. = 19.6°/25.0°, $ks = 37.9$, $\alpha_{95} = 6.8^\circ$, $N = 13$ (Fig. 8), indicating an inclination difference of only 2.7°. Although, high degree of inclination shallowing has been anticipated for clay-size sediments (Tan *et al*. 2002), only small difference of 2.7° in the sandstone and mudstone samples of this study can be considered as a striking observation.

Directional comparison between 10 pairs of sandstone and mudstone layers can also confirms similarity in inclinations. Mean directions obtained from 10 pairs of sandstone and mudstone layers are dec./inc. = 21.3°/23.4°, $ks = 35.8$, $\alpha_{95} = 8.2^\circ$, $N = 10$ and

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

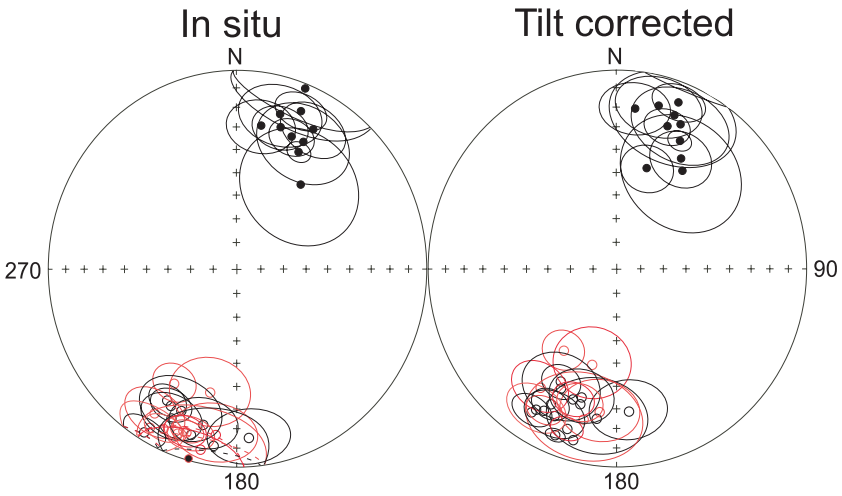


Figure 7. Equal-area projections of the site mean directions (circles) for high-temperature components with 95 per cent confidence limit before and after tilt-correction. All 30 reliable sites from the Lower Cretaceous Bailong and Cangxi formations are used in these projections. Solid and open symbols correspond to projection on lower and upper hemisphere, respectively.

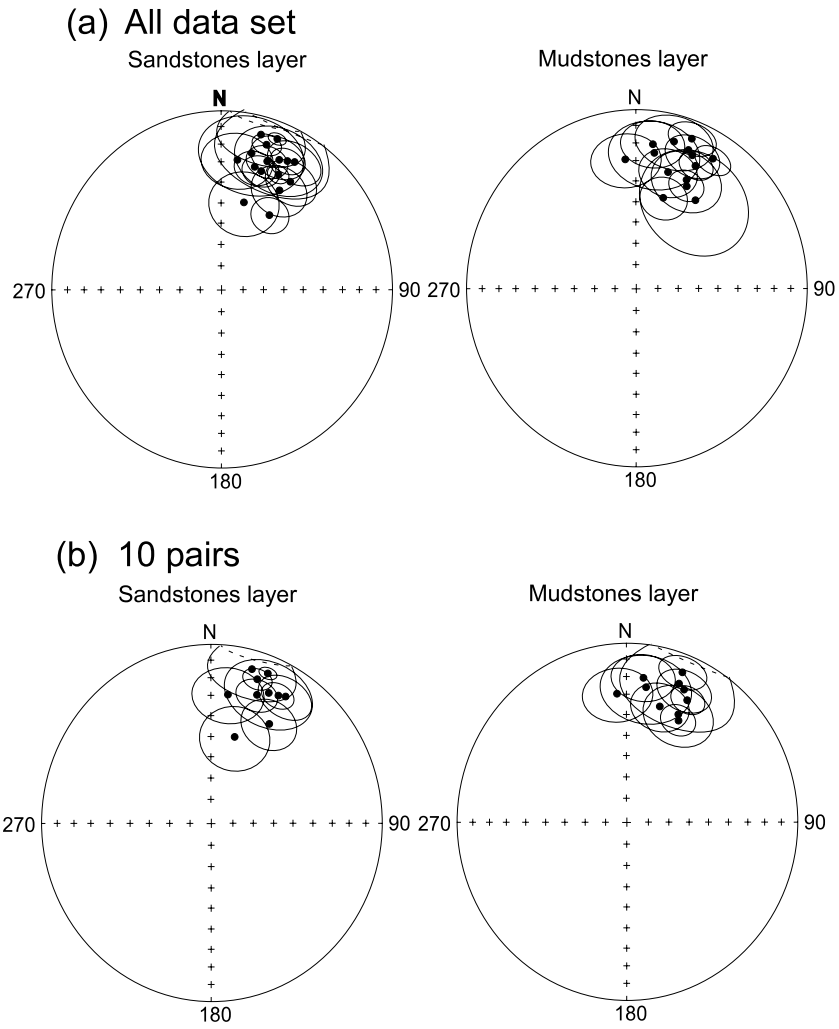


Figure 8. Comparison of the site mean directions (circles) between sandstone and mudstones sites on equal-area projection; (a) all data set and (b) 10 pairs of sandstone and mudstone layers. Reversed polarity directions are flipped into normal. Solid symbols correspond to projection on lower hemisphere.

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

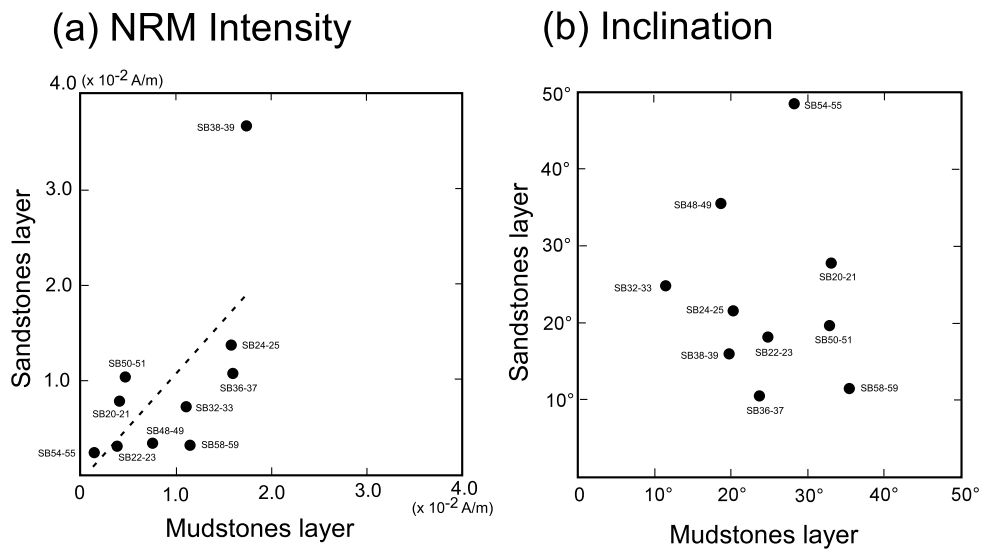


Figure 9. Comparison of the NRM intensities (a) and inclinations (b) between the paired sandstone and mudstone sites (10 pairs). Abscissa and ordinate show data from sandstone and mudstone layers, respectively. A best fit line based on linear regression for 10 pairs of the NRM intensities is drawn by dashed line: NRM intensity from sandstones = –0.07 + 1.13 × NRM intensity from mudstones. Correlation coefficient of these 10 pairs is 0.65 for NRM intensity and 0.09 for inclination.

dec./inc. = 17.4°/25.1°, $ks = 43.5$, $\alpha_{95} = 7.4°$, $N = 10$, respectively (Fig. 8). A difference in arithmetic means of inclinations is calculated as 1.4° ± 9.9°. On the basis of these observations, it is recognized that almost identical directions of the remanent magnetization (statistically) are preserved in the sandstone and mudstone layers.

Grain size dependence of inclination variation in the red beds is still considered as controversial issue. In this connection three different approaches have been recognized; (1) no inclination shallowing in coarse-grained sediments but shallowing in clay-sized sediments; (2) no inclination shallowing in fine-grain sediments but shallowing in coarse-grained sediments and (3) similar trend of inclination shallowing both in coarse and fine-grained sediments. According to their experimental work, Tan *et al*. (2002) have claimed that hematite bearing clayey sediments underwent a compaction related inclination shallowing of 17°–19°, while coarse-grained sediments received very little inclination shallowing. Similar type of behaviour has been reported by Hodych & Hayatsu (1988) on the basis of their palaeomagnetic results from Jurassic red beds. Contrary to these views some of the researchers (Tucker 1979; Stamatakos *et al*. 1995; Tan *et al*. 2007) have observed that fine-grained magnetic minerals, which carry magnetically hard component and distributed spectra of unblocking temperatures, have accurately recorded magnetic field directions without any inclination error. However, higher degree of inclination shallowing have been observed by them in the coarse-grained sediments, which carry magnetically soft NRM component and discrete unblocking spectra. An alternative observation from Tauxe & Kent (1984) have noticed no obvious grain size dependence on variation in inclination, although, both coarse and fine-grained hematite sediments were subjected to inclination shallowing.

According to this study no systematic bias in inclination is observed in 10 pairs of sandstone and mudstone layers, however, six sandstone layers from these pairs reveals steeper inclination as compared to mudstone layers. Averaging of palaeomagnetic directions from these 10 pairs provides nearly identical palaeomagnetic directions for sandstone and mudstone layers, however, their comparison with expected palaeomagnetic direction indicate an inclination shallowing of 29° for sandstones and 27° for mudstones.

6.2 Locality-wise variation of inclination shallowing

A trend of Inclination shallowing (flattening) observed through this study from Bazhong area (25.8° ± 4.0°) seems compatible with those reported from Ya'an (18.6° ± 9.2°) and Yichang (25.4 ± 6.2°) areas of the Sichuan Basin. However, red beds from Bazhong area have low degree of AMS ($1.028 < Pj < 1.140$) as compared to red beds from Ya'an and Yichang areas. This low degree of AMS is very well comparable to that reported from eastern coast of the SCB (red beds), where no inclination shallowing has been observed (Narumoto *et al*. 2006). Low degree of AMS, which indicate the existence of depositional fabrics in the sediments, has been linked to either zero or limited inclination shallowing (Gareces *et al*. 1996). These facts, thus, suggest that locality-wise changes is another factor to be considered for variation in inclination shallowing, although, depositional processes during sedimentation and compaction have been advocated as a major cause of inclination shallowing by Tauxe & Kent (2004) and Tan *et al*. (2007).

Very recently, locality-wise variation of inclination shallowing in central–southeast Asia have been assessed by Otofuji *et al*. (2010) using a compilation of palaeomagnetic data from Late Jurassic to Eocene red beds (Table 2). In this compilation (as shown in the-Figs 10a and b), magnitudes of inclination shallowing and respective latitudinal displacements have been evaluated with respect to palaeopoles (or palaeolatitudes) expected from the European APWPs of Besse & Courtillot (2002). Using the General Mapping Tools of Wessel & Smith (1991), contours of flattening and latitude displacement have been plotted by in-putting all the data sets. After assigning a null value to several localities (i.e. along the Pacific coastline and southern margin of the Himalayan mountain ranges), a solution using a tension parameter $T_1 = 0.4$ (Smith & Wessel 1990; Wessel & Smith 1991) have been applied to depict a first-order approximation for rotation.

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

Table 2. Late Jurassic to Eocene palaeomagnetic data available from areas around the eastern Himalayan syntaxis. Lon., longitude; Lat., latitude; Age: J, Jurassic; J2 Middle Jurassic, J3, Late Jurassic; K, Cretaceous, K1, Early Cretaceous; K2, Late Cretaceous, E1, Paleocene-Eocene. N (n), number of sites (samples) used for palaeomagnetic statistics. Dec. and Inc. are declination and inclination, respectively. α_{95} is a radii of the cone of 95 per cent confidence about the mean direction. Field test; positive and inconclusive (fold tests). Shallowing (flattening) and northward latitudinal displacement are evaluated by comparing the observed palaeomagnetic inclination (I_{obs}) and palaeolatitudes (λ_{obs}) with those expected from the Eurasian APWP (Besse & Courtillot 2002); Shallowing $= I_{ex} - I_{obs}$, northward latitudinal displacement $= \lambda_{ex} - \lambda_{obs}$. Uncertainty at 95 per cent confidence in shallowing and latitudinal displacement (Delta) is calculated by using a method of Demarest (1983).

Locality							Observed direction							
Name	Lat. (°N)	Lon. (°E)	Age	N	Rock type	Fold test	Dec. (°)	Inc. (°)	α_{95} (°)	Shallowing	Delta	Lat.displace	Delta	Reference
Hexi Corridor														
Yumen	39.9	97.7	K1	9	Red sandstones	Positive	18.9	61.7	5.7	–3.8	4.9	–4.3	6.5	Chen *et al*. (2002a)
Longshou	39.1	100.5	K1	36	Purple siltstones	Inconclusive	6.8	53.4	3.8	4.3	3.6	4.4	4.0	Dupont-Nivet *et al*. (2003)
Tarim Block														
Kuche River	42.1	83.2	K	21	Red beds	Positive	16.3	29.0	7.4	28.3	6.2	22.4	6.1	Tan *et al*. (2003)
Paicheng	41.8	82.0	J3-K1	6	Sandstones & mudstones	Positive	22.0	42.0	9.0	14.8	7.5	13.1	7.5	Li *et al*. (1988)
Kuche	41.6	83.5	K2	4	Redbeds	Positive	16.0	39.0	9.0	17.0	8.6	14.5	8.6	Li *et al*. (1988)
Tuoyun	40.2	75.3	K1	5	Redbeds	Positive	37.9	38.6	6.8	15.5	5.8	12.9	5.2	Gilder *et al*. (2003)
Tuoyun	40.2	75.3	E1	20	basalt & Redbeds	Positive	48.4	49.1	6.6	9.9	5.9	9.8	6.5	Huang *et al*. (2005)
Tuoyun	40.2	75.3	K1	13	Volcanics	Positive	31.9	49.9	10.6	4.2	8.7	3.9	9.8	Gilder *et al*. (2003)
Wuqia	39.5	75.0	K2	11	Redbeds	Positive	11.0	40.0	7.4	13.0		10.8		Chen *et al*. (1992)
Yingjisha	38.5	76.4	K2	6	Redbeds	Positive	7.6	37.1	9.9	15.0		12.0		Chen *et al*. (1992)
Hotan	37.2	79.0	E1	5	Redbeds	None	27.8	30.0	13.3	26.8	11.0	21.3	10.5	Gilder *et al*. (1996)
Puska	37.1	78.4	E1			Positive	4.3	24.3	8.4	32.3		24.5		Rumelhart *et al*. (1999)
Qaidam Block														
Sunan	39.0	99.6	K1	16	Redbeds	Positive	41.9	39.0	5.1	18.5		16.0		Frost *et al*. (1995)
Huatugou	38.5	90.8	J3	9	Red to grey-green sandstones	Positive	40.4	34.8	7.5	26.5	8.0	23.2	7.9	Chen *et al*. (2002b)
East Xining lower	36.6	101.9	K1	20	Redbeds	Positive	37.7	33.6	4.5	22.1		17.9		Dupont-Nivet *et al*. (2004
Shancheng	36.4	102.1	K2	15	Redbeds	Positive	30.1	33.9	9.6	19.4		15.3		Dupont-Nivet *et al*. (2004)
Xining-Lanzhoi	36.2	103.5	K1	12	Redbeds	Positive	45.0	43.7	4.3	11.9	3.9	10.6	4.2	Halim *et al*. (1998)
Lanzhou	36.0	103.5	K1	5	Redbeds	Positive	51.8	44.1	10.0	11.3		10.1		Frost *et al*. (1995)
Zeku	35.2	101.8	K1	19	Redbeds	Positive	17.4	53.9	4.5	0.4	4.1	0.4	4.7	Sun *et al*. (2006a)
Longsi A	34.9	104.4	K1	6	Redbeds	Positive	47.7	40.1	6.2	14.5		12.3		Dupont-Nivet *et al*. (2004)
Danchang	34.1	104.5	K1	18	Redbeds	Positive	35.8	50.9	3.0	2.9		2.7		Dupont-Nivet *et al*. (2004)
Kunlun Block														
Tula (west)	37.6	86.9	J-K	41	Redbeds	Positive	8.3	28.6	5.5	23.9		17.8		Dupont-Nivet *et al*. (2004)
Tula (east)	37.5	87.2	J-K	9	Redbeds	Positive	21.3	34.2	9.6	18.2		14.2		Dupont-Nivet *et al*. (2004)
Maqin	34.5	100.1	K1	13	Redbeds	Positive	359.8	42.5	10.3	10.8		9.2		Halim *et al*. (1998)
Qiangtang Block														
Aksaichin	35.0	79.7	K	7	Redbeds	Positive	1.3	21.8	6.6	26.9		18.3		Chen *et al*. (1993)
Quanshuihe (19)	34.8	79.9	K	1(5)	Siltstones	None	–8.5	–10.4	11.0	58.9		34.7		Otofuji *et al*. (1989)
Erdaogou	34.6	92.8	E1	12	Redbeds	Inconclusive	35	34	4.9	22.5		18.4		Lin and Watts (1988)
Erdaogou	34.6	92.4	E1	12	Redbeds	Inconclusive	9	14	7.2	42.4		29.9		Lin and Watts (1988)
Longmuco	34.5	80.4	K	(4)	Redbeds	None	12	22.2	12.8	26.0		17.7		Otofuji *et al*. (1989)
Fenghuoshar	34.4–34.7	92.7–92.9	E1	7	Redbeds	Positive	9.5	34.3	8.9	22.0		18.1		Halim *et al*. (1998)

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

Table 2. (*Continued.*)

Locality							Observed direction							
Name	Lat. (°N)	Lon. (°E)	Age	*N*	Rock type	Fold test	Dec. (°)	Inc. (°)	α_{95} (°)	Shallowing	Delta	Lat.displace	Delta	Reference
Bangonco (21)	33.5	79.9	J-K	1(4)	Shale	None	17.5	28.2	23.1	18.7		13.1		Otofuji *et al*. (1989)
Ritu (37)	33.3	79.7	K	1(5)	Granite	None	30.1	24.3	3.3	22.4		15.2		Otofuji *et al*. (1989)
Yanshiping	33.6	92.1	J2-3	20	Redbeds	Syn-folding	339.0	58.0	7.2	–0.6		–0.6		Lin and Watts (1988)
Xialaxiu	32.8	96.6	E	10	Redbeds	Positive	322.0	32.3	9.5	23.0		18.3		Cogne *et al*. (1999)
Markam	29.7	98.4	K1	12	Redbeds	Positive	57.7	50.2	10.9	–2.5		–2.1		Huang *et al*. (1992)
Markam	29.7	98.6	K	5	Redbeds	Inconclusive	48.2	49.0	8.8	–3.7	8.5	–3.1	9.0	Otofuji *et al*. (1990)
Markam	29.7	98.7	K1-2	11	Redbeds	Positive	39.7	51.6	8.5	–6.3		–5.4		Huang *et al*. (1992)
Lhasa Block (northern part)														
Qelico	31.7	90.9	K1-2	4	Andesite	Inconclusive	347.0	36.0	16.5	10.4		7.8		Lin and Watts (1988)
Naqu	31.5	91.0	K1-2	9	Andesite	Inconclusive	358.0	35.0	6.0	11.2		8.2		Lin and Watts (1988)
Anduo	31.0	92.0	K1	6	Redbeds	Positive	338.7	25.4	8.9	22.3	7.4	15.4	5.5	Achache *et al*. (1984)
Basu	30.1	96.9	J2	8	Redbeds	Positive	353.4	15.1	14.1	40.3		28.2		Otofuji *et al*. (2007)
Basu	30.1	96.9	127,115Ma	2	Granite	None	343.7	12.0		35.8		22.8		Otofuji *et al*. (1990)
Lhasa Block (southern part)														
Jiangba (4)	32.5	80.4	K	1(6)	Diorite	None	294.7	17.5	6.0	28.3		18.3		Otofuji *et al*. (1989)
Shiquanhe (6)	32.5	80.0	E	1(6)	Red shale	None	15.4	22.7	12.0	29.6		21.1		Otofuji *et al*. (1989)
Shiquanhe	32.7	80.2	K	3	Limestone	Positive	9.8	23.5	23.0	22.5		15.1		Chen *et al*. (1993)
Takena (southern region)	30.0	91.0	K1	8	Redbeds	Inconclusive	354.3	22.6	8.3	23.7	6.9	15.9	6.6	Achache *et al*. (1984)
Linzhou	30.0	91.1	E	10	Tuffs	Positive	359.5	51.8	5.2	–0.2		–0.2		Tan *et al*. (2010)
Lingzizong	30.0	91.0	E1	8	Volcanics. Intrusive	Positive	350.9	25.5	11.0	26.1		18.8		Achache *et al*. (1984)
Takena (Linzhou)	29.9	91.2	K2	8	Redbeds	Positive	357.0	15.0	6.7	29.3		18.4		Lin and Watts (1988)
Linzhou	29.9	91.2	K2	43	Redbeds	Positive	350.2	23.5	2.5	20.8		13.7		Tan *et al*. (2010)
Linzhou	29.9	91.2	K2	21	Lava flows	Positive	22.6	41.9	4.4	2.4		1.8		Tan *et al*. (2010)
Quxu (T28)	29.5	91.0	E1	1(5)	Granodiorite	None	267.2	45.4	16.5	5.6		4.8		Otofuji *et al*. (1991)
Quxu (T29)	29.5	91.0	E1	1(5)	Granite	None	312.4	31.0	12.2	20.0		15.0		Otofuji *et al*. (1991)
Yarlung Zangbo suture zone (Xigaze Forearc Basin)														
Donglha	29.1	88.4	K1	8	Chert	Inconclusive	306.7	–5.4	6.7	50.0		28.9		Abrajevitch *et al*. (2005)
Qunrang	29.2	89.0	K1	17	Chert, clastic rocks	Inconclusive	297.1	15.6	5.1	29.3		18.5		Abrajevitch *et al*. (2005)
Cheujeon (14R)	29.2	89.5	K1	1(14)	Radiolarites	None	279	28	7	17.0		11.7		Pozzi *et al*. (1984)
Cheujeon(14B)	29.2	89.5	K1	1(10)	Basalts	None	239	24	15	21.0		14.0		Pozzi *et al*. (1984)
Xigaze airport (18)	29.2	89.5	K1	1(9)	Limestone	None	270	23	15	22.0		14.6		Pozzi *et al*. (1984)
Yamu (35)	29.4	87.7	K2-E1	1(5)	Sandstones	None	293.1	–2.6	27.9	53.0		32.4		Otofuji *et al*. (1989)
Laabgcuo (29)	30.7	81.4	K	1(3)	Gabbro	None	298.9	6.8	22.8	36.9		22.1		Otofuji *et al*. (1989)
Mentu (25)	31.3	80.9	E1	1(5)	Sandstones	None	298.5	52.0	18.9	–0.8		–0.7		Otofuji *et al*. (1989)
Chuan Dian Fragment														
Xichang	27.9	102.3	K1	33	Redbeds	Positive	3.7	41.5	3.4	5.1	3.3	4.0	3.4	Tamai *et al*. (2004)
Huili	26.0	102.5	K1	7	Redbeds	Inconclusive	22.1	37.1	3.8	7.3	3.6	5.3	3.9	Huang and Opdyke (1992)
Huili	26.4	102.3	E	7	Redbeds	Inconclusive	359	13.7	12.1	35.8	10.1	23.4	9.7	Huang and Opdyke (1992)
Huili	26.5	102.4	K2	18	Redbeds	Positive	8.1	38.8	6.6	3.1	7.1	2.3	7.4	Huang and Opdyke (1992)
Yongren	26.1	101.7	E	16	Redbeds	Positive	17.2	26.6	5.8	22.5	5.4	15.9	4.8	Yoshioka *et al*. (2003)
Dayao	25.7	101.3	E	10	Redbeds	Positive	16.5	31.1	4.8	17.5	4.7	12.8	4.5	Yoshioka *et al*. (2003)
Zhupeng	25.9	101.8	K2,E	15	Redbeds	Positive	25.4	34.3	3.4	14.6	3.8	11.0	3.4	Otofuji *et al*. (1998)
Bailu	25.7	102.1	K2,E	6	Redbeds	Positive	30.7	39.0	10.6	9.7	8.9	7.6	8.8	Otofuji *et al*. (1998)

© 2011 The Authors, *GJI*, **186**, 471–491
Geophysical Journal International © 2011 RAS

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

Table 2. (*Continued.*)

Locality							Observed direction							
Name	Lat. (°N)	Lon. (°E)	Age	*N*	Rock type	Fold test	Dec. (°)	Inc. (°)	α_{95} (°)	Shallowing	Delta	Lat.displace	Delta	Reference
Chuxiong W	25.3	101.3	K1	6	Redbeds	Positive	30.6	48.5	11.1	–5.3	9.1	–4.3	9.9	Funahara *et al*. (1992;
Chuxiong E	25.3	101.8	K1	4	Redbeds	Positive	47.8	36.5	11.8	6.8	9.6	4.9	10.7	Funahara *et al*. (1992;
Shan-Thai Block														
Yunlong	25.8	99.4	K2	20	Redbeds	Positive	40.2	49.9	3.9	–9.5	5.7	–7.6	5.9	Sato *et al*. (1999)
Yunlong	25.8	99.4	K2	29	Redbeds	Positive	38.3	50.7	3.4	–10.3	5.4	–8.4	5.7	Yang *et al*. (2001)
Xiaguan	25.6	100.2	K2	9	Redbeds	Positive	6.9	47.7	8.6	–7.4	8.3	–5.8	9.3	Huang and Opdyke (1993)
Weishan	25.4	100.2	J3	5	Redbeds	Inconclusive	7.3	25.3	10.4	26.2	9.9	18.9	9.9	Huang and Opdyke (1993)
Yongping	25.5	99.5	K1	12	Redbeds	Positive	42.0	51.1	15.7	–8.2	12.7	–6.8	16.6	Funahara *et al*. (1993;
Jingdong	24.5	100.8	K1-2	13	Redbeds	Positive	8.3	48.8	7.7	–9.9	7.8	–7.8	8.5	Tanaka *et al*. (2008)
Luxi	24.3	98.4	J2	6	Redbeds	Inconclusive	99.7	35.2	11.3	14.7	10.5	11.3	11.1	Huang and Opdyke (1993)
Zhengyuar	24.1	101.1	K1-2	7	Redbeds	Positive	61.8	46.1	8.1	–7.7	8.0	–5.8	8.0	Tanaka *et al*. (2008)
Zhengyuar	24.1	101.1	K1-2	4	Redbeds	Positive	144.2	49.4	6.4	–11.0	7.0	–8.6	7.8	Tanaka *et al*. (2008)
Jinggu	23.6	100.5	J2	10	Redbeds	Inconclusive	83.3	36.8	5.4	12.9	6.8	10.0	6.3	Huang and Opdyke (1993)
Jinggu	23.4	100.4	K1	–	Redbeds	Inconclusive	84.4	39.6	17.8	0.9		0.6		Chen *et al*. (1995)
Jinggu	23.4	100.5	K2	7	Redbeds	Positive	115.8	36.0	6.3	1.3		0.9		Chen *et al*. (1995)
Jinggu	23.4	100.9	K2	8	Redbeds	Inconclusive	79.4	43.3	9.1	–5.9	8.7	–4.3	8.5	Huang and Opdyke (1993)
Pu'er	23.0	101.0	K1-2	25	Redbeds	Positive	59.9	45.2	5.1	–8.4	6.2	–6.2	6.5	Sato *et al*. (2007)
Mengla	21.6	101.4	K2	10	Redbeds	Inconclusive	60.8	37.8	7.6	–3.0	7.7	–2.0	8.1	Huang and Opdyke (1993)
South Mengla	21.4	101.6	K1-2	14	Redbeds	Positive	51.2	46.4	5.6	–11.8	6.5	–8.7	6.8	Tanaka *et al*. (2008)
Phong Saly	21.6	101.9	J3-K1	19	Redbeds	Positive	29.8	32.7	9.1	5.7	7.5	3.8	6.4	Takemoto *et al*. (2009)
Kalaw	20.7	96.5	J-K	13	Redbeds	Positive	44.7	23.4	6.1	9.0	6.8	5.4	6.1	Richter and Fuller (1996)
Nan	19.2	101.0	J1-3	11	Redbeds	Positive	32.2	33.3	12.2	11.5	11.1	8.2	10.7	Aihara *et al*. (2007)
Indochina Block														
Lai Chau	22.3	103.4	K3	5	Redbeds	Positive	12.2	40.1	4.7	–3.8	6.0	–2.7	6.2	Takemoto *et al*. (2005)
Yen Chau	21.0	104.4	K3	8	Redbeds	Positive	3.2	26.7	12.9	7.9	11.3	4.9	9.8	Takemoto *et al*. (2005)
Borikhanxav	18.5	103.8	J3-K1	18	Redbeds	Positive	41.8	46.6	8.0	–12.0	6.7	–8.9	7.2	Takemoto *et al*. (2009)
Amphoe Bung Kuan	18.2	103.9	K2	14	Redbeds	Inconclusive	31.8	28.7	3.5	1.4	5.5	0.9	5.5	Charusiri *et al*. (2006)
Nam Nao	16.5	103.0	K1	10	Redbeds	Inconclusive	28.1	40.5	2.4	–9.2	2.7	–6.2	2.7	Yang and Besse (1993)
Nam Nao	16.5	103.0	J3	10	Redbeds	Inconclusive	26.6	37.7	2.6	4.3	5.7	3.1	5.6	Yang and Besse (1993)
Muan Sakon Nakhor	16.5–17.2	102.5–104.1	K2	8	Redbeds	Inconclusive	31.4	27.1	9.4	1.0	8.9	0.6	8.9	Charusiri *et al*. (2006)
Muan Sakon Nakhor	16.5–17.2	102.5–104.1	K1	4	Redbeds	Positive	31.8	38.3	5.7	–6.1	4.9	–4.1	4.9	Charusiri *et al*. (2006)
Muan Phir	16.5	106.0	J	17	Redbeds	Positive	34.2	34.1	4.6	8.8	6.4	6.2	6.2	Takemoto *et al*. (2009)
Da Lat	10.4–12.9	105.0–109.4	K	21	Ryolite, Ryolite tuff, granite,sil	None	14.5	33.3	6.3	–13.1	6.9	–7.7	6.7	Chi and Dorobek (2004)

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

Table 2. (*Continued.*)

Locality							Observed direction							
Name	Lat. (°N)	Lon. (°E)	Age	*N*	Rock type	Fold test	Dec. (°)	Inc. (°)	α_{95} (°)	Shallowing	Delta	Lat.displace	Delta	Reference
South China Block (SCB)														
Cretaceous-Paleocene														
Bazhong														
	32.1	106.8	K1	30	Redbeds	positive	20.9	26.5	4.4	25.8	4.0	18.9	3.5	This study
Nanjing	32.0	119.0	K2	10	Purple sandstones	Positive	13.7	58.0	7.6	–6.6	7.7	–6.6	9.5	Kent *et al*. (1986)
Anhui	30.8	118.2	K2-E	7	Purple siltstones	Positive	8.4	47.7	6.4	8.6	5.8	8.1	12.0	Gilder *et al*. (1999)
Yichang	30.7	111.7	K2	26	Redbeds	Positive	3.8	23.3	5.1	25.4	6.2	17.5	5.8	Narumoto *et al*. (2006)
Xinjin	30.4	103.8	K2	4	Redbeds	inconclusive	19.2	39.2	12.1	7.8		6.0		Enkin *et al*. (1991b)
Ya'an	30.1	103.0	K1	9	Redbeds	Positive	2.1	30.7	11.3	18.6	9.2	13.6	9.1	Otofuji *et al*. (1990)
Feixianguar	30.0	102.9	J3-K2	26	Redbeds	Positive	2.0	34.2	3.6	12.2		8.9		Enkin *et al*. (1991b)
Guanyir	29.1	104.6	J3-K2	14	Redbeds	Inconclusive	15.8	32.7	7.8	12.9		9.3		Enkin *et al*. (1991b)
Zhejiang	28.4–29.5	119.9–121.0	K1	19	Redbeds	Positive	9.6	47.6	5.5	4.4	4.8	3.9	5.8	Morinaga and Liu (2004)
Zhejiang	28.4–29.1	119.6–120.0	K2	19	Redbeds	Positive	10.4	45.5	5.7	2.7	6.6	2.3	7.0	Morinaga and Liu (2004)
Yuanma	27.7–28.6	109.6–110.7	K1	13	Redbeds	Positive	12.0	48.5	6.4	0.1	5.5	0.1	5.6	Zhu *et al*. (2006)
Yuanma	27.7–28.6	109.6–110.7	K2	17	Redbeds	Positive	6.2	50.3	4.0	–5.1	5.7	–4.3	6.2	Zhu *et al*. (2006)
Jishui	27.1	115.1	K2	10	Redbeds	Positive	355.7	34.8	6.3	10.3	6.9	7.5	6.6	Wang and Yang (2007)
Hengyuang	26.9	112.9	E1	22	Redbeds	Reversal test	358.9	35.4	5.0	16.3	4.8	12.8	4.4	Sun *et al*. (2006b)
Hengyuang	26.9	112.9	K2	26	Redbeds	Positive	15.6	29.9	5.7	14.5	6.6	10.1	6.0	Sun *et al*. (2006b)
Jiangxi	26.9	115.3	K1	23	Redbeds	51.2 per cent untiltted	12.9	38.6	3.4	10.0	3.3	7.8	3.3	Tsuneki *et al*. (2009)
Ganzhou	25.9	115.0	K2	14	Redbeds	Positive	15.6	35.6	5.5	8.0	6.5	5.7	6.3	Wang and Yang (2007)
Chijiang	25.5	114.5	Paleocene	73	Redbeds	Stratigraphy	12.8	23.5	4.0	26.9	4.2	18.9	3.9	Clyde *et al*. (2008)
Fujian	25.1–26.4	116.4–117.8	K2	22	Redbeds	Positive	11.5	44.2	5.3	–0.1	6.3	–0.1	6.6	Morinaga and Liu (2004)
Guangdong	24.1–24.2	115.2–115.8	K2	9	Redbeds	Positive	10.7	44.0	8.0	–2.8	8.0	–2.1	7.8	Morinaga and Liu (2004)
Nanning	22.7	108.7	K1	8	Redbeds	Positive	352.8	38.2	5.5	3.6	4.8	2.6	5.8	Gilder *et al*. (1993)
Hong Kong	22.2	114.1	K	12	Dyke	Positive	23.8	34.4	5.3	4.0	6.3	2.7	8.5	Li *et al*. (2005)
Hainan	19.0	109.2	K1	6	Redbeds	Positive	4.3	41.8	9.4	–4.9	7.8	–3.5	8.1	Li *et al*. (1995)
Hainan	19.0	109.2	K1	17	Redbeds	54 per cent untilted	4.3	48.4	5.4	–11.5	4.7	–8.8	4.0	Liu and Morinaga (1999)
Late Jurassic														
Wangchang	32.1	106.2	J3	6	Redbeds	Positive	18.4	29.3	8.5	30.0	8.6	24.4	8.1	Bai *et al*. (1998)
Bazhong	31.8	106.7	J3	11	Redbeds	Positive	20.0	28.8	5.8	30.3	7.0	24.5	7.7	Yokoyama *et al*. (1991)
Jiangyang	30.4	104.5	J3	24	Redbeds	Positive	25.9	26.7	4.7	30.7	6.5	23.9	6.3	Yokoyama *et al*. (2001)

Notes. Europe (Besse & Courtillot 2002). 50 Ma (E1). 80 Ma (K2). 120 Ma (K1). 150 Ma (J, J1–J3).

© 2011 The Authors, *GJI*, **186**, 471–491
Geophysical Journal International © 2011 RAS

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

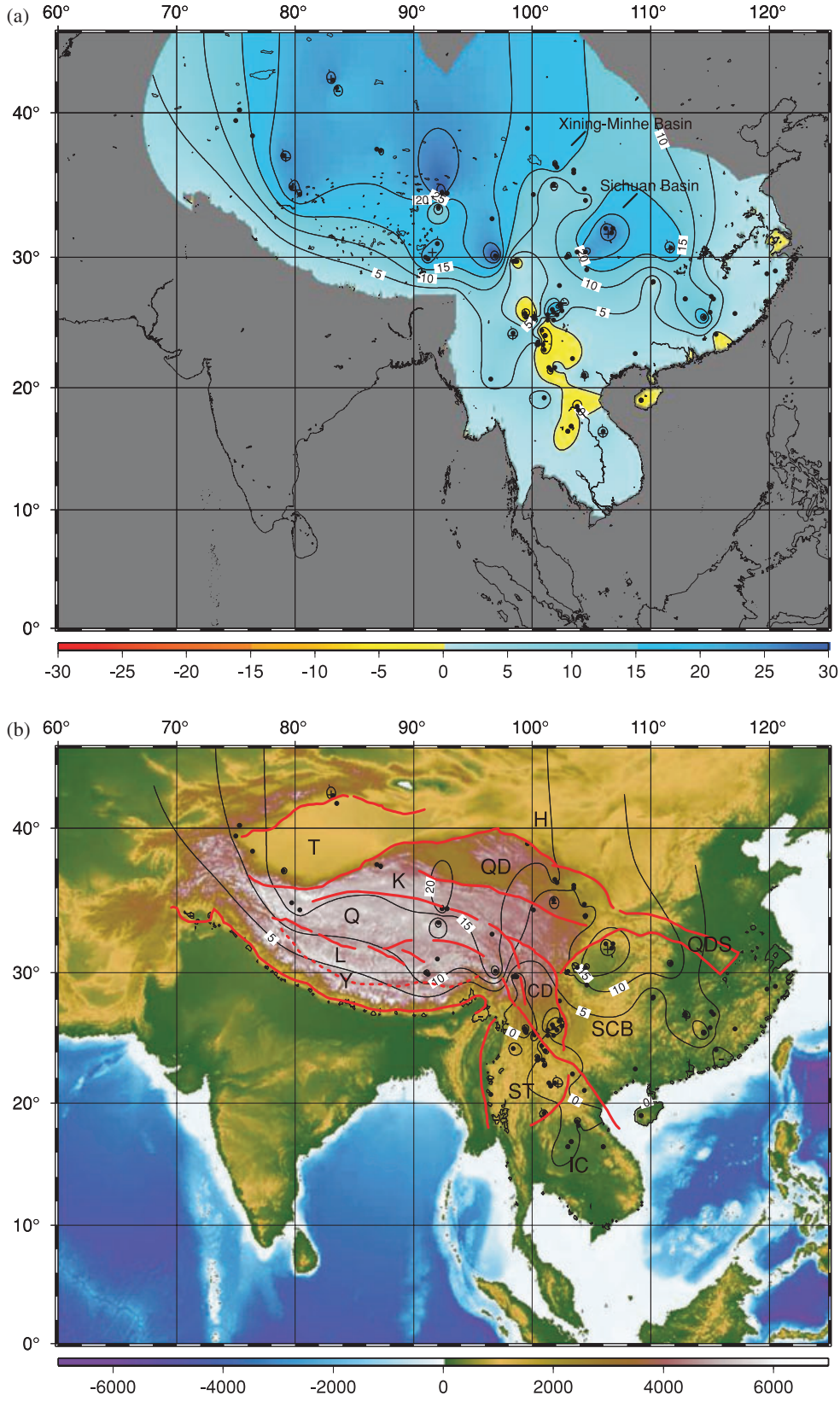


Figure 10. Maps of inclination shallowing and latitudinal displacement. (a) Contours showing an inclination shallowing (flattening). These contours are plotted at 5° interval by inputting inclination shallowing data using the General Mapping Tools of Wessel & Smith (1991). (b) Contours of northward latitudinal displacement are drawn on the topographic map of eastern Himalayan syntaxis and surrounding areas. Tectonic zonations are same as explained in Fig. 1.

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

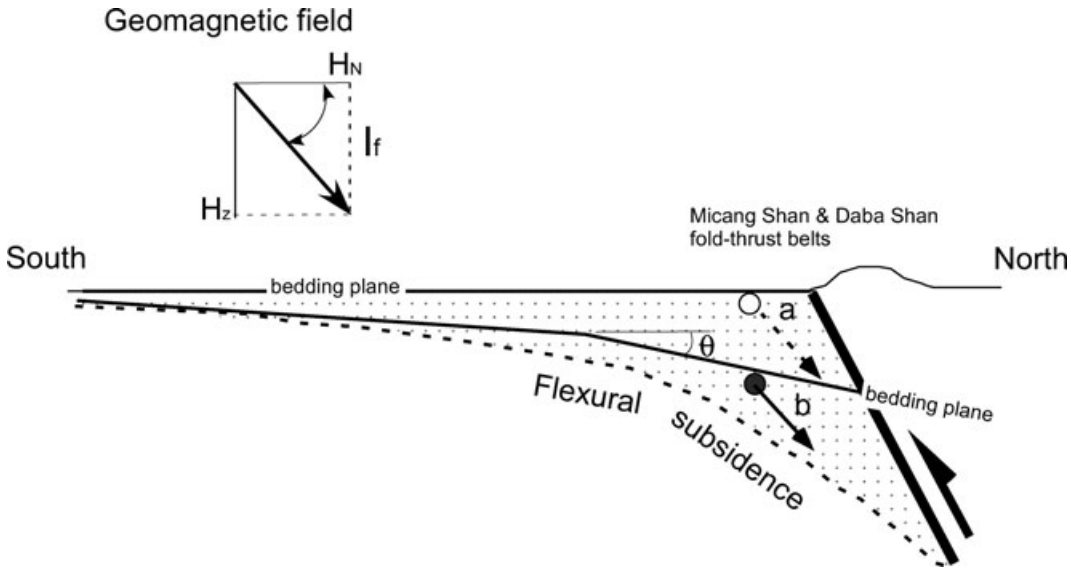


Figure 11. Inclination shallowing model in north-bounded flexural subsidence basin with typical wedged-shaped geometry. Occurrence of flexural subsidence of the NW Sichuan Basin was probably caused by activities along the east–west striking Micang Shan and Daba Shan fold-thrust belts. Dotted arrow represent an un-locked position of the remanent magnetization at 'a'. Solid arrow at 'b' is the locked position of the remanent magnetization, acquired at a time when sedimentary layer was already dipping northward by θ degrees. An observed inclination (I_o) with respect to bedding plane at 'b' is presented as $I_o = I_f - \theta$, where I_f denote an inclination of the geomagnetic field. Hz and H_N are the vertical and northward components of the palaeomagnetic field.

As shown in Fig. 10(a), features of inclination shallowing in central and southeast Asia are described by the following four aspects. (1) a pervasive extended region of more than 20° inclination shallowing in central Asia, (2) very low degree of inclination shallowing in coastal area of the SCB, (3) locally high degree of inclination shallowing in the Sichuan Basin (western part of the SCB) and in the Xining-Minhe Basin (the Qaidam Block) and (4) area of steep inclinations in the Shan-Thai Block of Indochina Peninsula. All these factors, thus, identify that magnitude of inclination in different parts of central and east Asia is not uniformly/systematically distributed but varies on locality basis (from one block to another).

Steep inclination in the Shan-Thai Block can be ascribed to its southward displacement after the Indian Plate collision with Asia (Fig. 10b). In addition to geological evidences from the Ailaoshan-Red River fault zone, several numerical and laboratory experiments have confirm a postulated southward displacement of the Shan-Thai Block (Tapponnier *et al*. 1982; Houseman & England 1993; Leloup, *et al*. 1995; Royden *et al*. 2008). Recent GPS observations also suggest southward displacement of this block (Gan *et al*. 2007).

6.3 Origin of locality-wise variation in inclination shallowing

Observation of almost zero inclination shallowing in the eastern part of the SCB provides a clue to understand the origin of this phenomenon (Fig. 10a). During the Mesozoic, the NE-striking grabens/basins were formed in the eastern part of the SCB as a result of large-scale extensional tectonics and resulted crustal thinning (Li 2000; Shu *et al*. 2009). On the basis of its topographic features with moderate undulation, this region has been referred as 'Basin and Range Province of Southeast China' (Li 2000). Palaeomagnetic studies of the Late Cretaceous to Palaeogene red beds from these basins, located along the east coast of the SCB, revealed no inclination shallowing (Gilder *et al*. 1993; Morinaga & Liu 2004; Zhu *et al*. 2006; Tsuneki *et al*. 2009). Mass bedding sequences were developed in the shape of Basin and Range Province during Late Cretaceous to Palaeogene, indicating an existence of stable and protected environmental conditions at the time of sedimentary deposition, such as deep-lake centres (Shu *et al*. 2009). The extensional tectonic regime with low topography can yield no shallow inclination.

The NW Sichuan Basin, where inclination shallowing have been observed, were formed in a quite different tectonic setting as compared to basin and range topography. Formation of this basin has been linked to Mesozoic tectonics of the east–west striking Qinling orogeny, which was instigated by a collision between the south Qinling and the SCB (Meng & Zhang 2000). As a foreland basin, the NW Sichuan Basin experienced flexural subsidence due to continued penetration of the SCB into the Qinling-Dabie suture zone (Meng *et al*. 2005). Activities along the east–west striking Micang Shan and Daba Shan fold-thrust belts played a major role in the initiation of flexural subsidence, which resulted in the formation of typical wedge-shaped geometry in the Early Cretaceous (Fig. 11).

Swift development of north-bounded flexural subsidence basin was probably responsible for the culmination of shallow inclination (Fig. 11). It is assumed that north-dipping stratum precedes an acquisition of remanent magntization during the formation of the foreland basin, suggesting that stratum is not always a record of palaeo-horizontal plane when sediments get magnetized. Assuming that direction of the palaeomagnetic field oriented northwards and the remanent magnetization achieved complete locking at a time when sedimentary layer was dipping northward by θ (in °), the observed inclination (I_o) and inclination shallowing ($I_f - I_o$) will be characterized by $I_o = I_f - \theta$ and θ, respectively, provided that inclination of the palaeomagnetic filed was represented by I_f ($I_f = \tan^{-1} Hz/H_N$, where Hz and H_N are vertical and northwards components of palaeomagnetic field, respectively). Alternatively, if westward bounded flexural subsidence basin developed at a time when the sediment layer dipped westward, the observed inclination could then be calculated as $\tan I_o = Hz \cos\theta/[H_N{}^2 + (Hz \sin\theta)^2]^{1/2}$. Comparison between these two phenomenons provides a clue that inclination shallowing in northward bounded basin is

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

could be significantly larger than that of the westward bounded basin.

These explanatory notes can be taken in support of shallow inclination, which is conspicuous on the southern flanks of high mountain ranges. Discordant shallow inclinations have been reported from the Cenozoic rocks of northern Qaidam Basin (Lu & Xiong 2009) and the Cretaceous red beds of the Xining-Minhe Basin. These basins were formed as foreland basins from Middle Jurassic to Cretaceous when their northern parts were bounded by reverse/thrust faulting along the east–west striking Qilian Shan (Yin & Harrison 2000; Ritts & Biffi 2001; Tapponnier *et al*. 2001; Roger *et al*. 2003; Yin *et al*. 2007). The Neogene red beds of Yaha section (Tarim Basin), in which an inclination shallowing has been detected, was also formed in a foreland basin environment on southern flank of the Tianshan Mountain (Charreau *et al*. 2005, 2006). It should be noted that no sign of inclination shallowing has been found on the northern flank of Qilian mountain ranges (Chen *et al*. 2002a; Dupont-Nivet *et al*. 2003). Based on these discussions, it can be claimed that the Mesozoic to Cenozoic inclination shallowing in east Asia originated in the north-bound foreland basins surrounded by east–west trending high mountain ranges.

According to numerical experiments conducted by Jezek & Gider (2006), high topographic relief around the sedimentary basins also enhances a chance of inclination shallowing, as faster sedimentation rate can help increase inclination shallowing compared to slow sedimentation. High-elevated terrains induces mass transportation of sediments to adjacent flexural subsidence basins through water and wind, which results in high sedimentation rate/or high loading rate in the basins. High sedimentation rate is reported for the Early Cretaceous red beds in the Xining-Minhe Basin, where large degree of inclination shallowing has been observed. Average rate of sedimentation for this particular period has been determined at 25 m Myr^{-1}, which is ten times higher than that occurred during the Late Cretaceous to Early Palaeogene period (Horton *et al*. 2004).

7 CONCLUSION

With the help of this study, variation of inclination shallowing is examined by lithological as well as locality scale.

1. The Early Cretaceous red beds from Bazhong area (32.1°N, 106.7°E) carry a primary palaeomagnetic direction. Following the passage of both fold and reversal tests, a tilt-corrected mean direction of dec./inc. = 20.9°/26.5° ($ks = 37.2$, $\alpha_{95} = 4.4°$, $N = 30$) is estimated.

2. Comparison with the 120 Ma pole expected from European APWPs suggests an inclination shallowing by $25.8° \pm 4.0°$.

3. 10 pairs of successive sandstone and mudstone layers provided almost identical inclination values at 95 per cent confidence limit ($I_{\text{sandsones}} = 23.3° \pm 8.3°$ and $I_{\text{mudstones}} = 24.7° \pm 5.4°$), indicating that NRMs in these layers were subjected to similar degree of inclination shallowing.

4. Location-wise variation in inclination shallowing is observed in the Late Jurassic to Eocene palaeomagnetic results reported from the red beds of east Asia, where no inclination shallowing is observed along the coastal area of the SCB, but large imprints of inclination shallowing have been detected locally in the western part of the SCB (Sichuan Basin).

5. No inclination shallowing has been observed in the 'Basin and Range' type topography, where steady depositional environment (such as deep-lake centres) had probably blocked the occurrence of inclination shallowing.

6. Inclinations shallowing in the red beds of sedimentary basins were most likely caused by flexural subsidence, which were bounded by high altitude orogenic belts in the north. It seems a reasonable explanation that acquisition of remanent magnetization occurred at a time when sedimentary layers were already dipping towards north by $\theta°$, and inclination shallowing of significant degrees was occurred. High rate of sedimentation and swift load movement in highly erodible mountainous environments could have caused an enhancement of inclination shallowing.

ACKNOWLEDGMENTS

We are thankful to two anonymous reviewers for their constructive comments on the manuscript. This research was partly supported by Toyota Foundation, Grant-in aid (Nos. 14403010, 18403012 and 22403012) from the Japanese Ministry of Education, Culture, Sports, Science and Technology (MEXT). Support provided by Global Centers of Excellence (GCOE) Program 'Foundation of International Center for Planetary Science' of the MEXT is greatly appreciated.

REFERENCES

Abrajevitch, A.V., Ali, J.R., Aitchison, J.C., Badengzhu, Davis, A.M., Liu, J., & Ziabrev, S.V., 2005. Neotethys and the India-Asia collision: insights from a palaeomagnetic study of the Dazhuqu ophiolite, southern Tibet. *Earth planet. Sci. Lett.,* **233,** 87–102.

Achache, J., Courtillot, V. & Xiu, Z.Y., 1984. Paleogeographic and tectonic evolution of southern Tibet since Middle Cretaceous time; new paleomagnetic data and synthesis. *J. geophys. Res.,* **89,** 10 311–10 339.

Aihara, K. *et al.*, 2007. Internal deformation of the Shan-Thai block inferred from Paleomagnetism of Jurassic sedimentary rocks in Northern Thailand, *J. Asian Earth Sci.,* **30,** 530–541.

Bai, L., Zhu, R., Wu, H. & Guo, B., 1998. Paleomagnetism of the Late Jurassic northern Sichuan basin and preliminary study on the true polar wander, *Acta geophys. Sin.,* **41,** 324–331.

Bazhenov, M.L. & Mikolaichuk, A.V., 2002. Paleomagnetism of Paleogene basalts from the Tienshan, Kyrgystan: rigid Eurasia and dipole geomagnetic field, *Earth planet. Sci. Lett.,* **195,** 155–166.

Besse, J. & Courtillot, V., 2002. Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr, *J. geophys. Res.,* **107,** 2300, doi:10.1029/2000JB000050.

Bureau of Geology and Mineral Resources of Sichuan Province, 1991. *Regional Geology of Sichuan Province, Geological Memoirs,* Series 1, Number 23, pp. 730, Geol Publishing House, Beijing. (BGMRSP)

Charreau, J. *et al.*, 2005. Magnetostoratigraphy and rock magnetism of the Neogene Kuitun He section (northwest China): implications for Late Cenozoic uplift of the Tianshan mountains, *Earth planet. Sci. Lett.,* **230,** 177–192.

Charreau, J. *et al.*, 2006. Magnetostoratigraphy of Yaha section, Tarim Basin (China): 11 Ma acceleration in erosion and uplift of the Tianshan Mountains, *Geology,* **34,** 181–184.

Charusiri, P., Imsamut, S., Zhuang, Z., Ampaiwan, T. & Xu, X., 2006. Paleomagnetism of the earliest Cretaceous to early late Cretaceous sandstones, Khorat Group, Northeast Thailand: implications for tectonic plate movement of the Indochina block, *Gondwana Res.,* **9,** 310–325.

Chauvin, A., Perroud, H. & Bazhenov, M.L., 1996. Anomalous low palaeomagnetic inclinations from Oligocene—lower Miocene red beds of the south-west Tien Shan, Central Asia, *Geophys. J. Int.,* **126,** 303–313.

Chen, H., Dobson, J., Heller, F. & Hao, J., 1995. Paleomagnetic evidence for clockwise rotation of the Simao region since the Cretaceous: a consequence of India-Asia collision, *Earth planet. Sci. Lett.,* **134,** 203–217.

Chen, Y., Cogné, J-P. & Courtillot, V., 1992. New Cretaceous paleomagnetic poles from the Tarim Basin, Northwestern China, *Earth planet. Sci. Lett.,* **114,** 17–38.

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

Chen, Y., Cogné, J-P., Courtillot, V., Tapponnier, P. & Zhu, X.Y., 1993. Cretaceous paleomagnetic results from western Tibet and tectonic implications, *J. geophys. Res.,* **98,** 17 981–17 999.

Chen, Y., Wu, H., Courtillot, V. & Gilder, S., 2002a. Large N-S convergence at the northern edge of the Tibetan plateau? New Early Cretaceous paleomagnetic data from Hexi Corridor, NW China. *Earth planet. Sci. Lett.,* **201,** 293–307.

Chen, Y., Gilder, S., Halim, N., Cogné, J-P. & Courtillot, V., 2002b. New paleomagnetic constraints on central Asian kinematics: displacement along the Altyn Tagh fault and rotation of the Qaidam Basin, *Tectonics,* **21**(5), 1042, doi:10.1029/2001TC901030.

Chi, C.T. & Dorobek, S.L., 2004. Cretaceous palaeomagnetism of Indochina and surrounding regions: Cenozoic tectonic implications, *Geol. Soc. London, Spec. Pub.*, **226,** 273–287.

Clyde, W.C. *et al*., 2008. An integrated stratigraphic record from the Paleocene of the Chinjang Basin, Jiangxi Province (China): implications for mammalian turnover and Asia block rotations, *Earth planet. Sci. Lett.,* **269,** 554–564.

Cogné, J.P. 2003. PaleoMac: a Macintosh^{TM} application for treating paleomagnetic data and making plate reconstructions, *Geochem. Geophys. Geosyst.* **4,** 1007, doi:10.1029/2001GC000227.

Cogné, J.P., Halim, N., Chen, Y. & Courtillot, V., 1999. Resolving the problem of shallow magnetizations of Tertiary age in Asia: insights from paleomagnetic data from the Qiangtang, Kunlun, and Qaidam blocks (Tibet, China), and a new hypothesis, *J. geophys. Res.,* **104,** 17 715–17 734.

Dekkers, M.J. & Linssen, J.H., 1989. Rockmagnetic properties of fine-grained natural low-temperature haematitie with reference to remanence acquisition mechanism in red beds, *Geophys. J. Int.,* **99,** 1–18.

Demarest, H.H., 1983. Error analysis of the determination of tectonic rotation from paleomagnetic data, *J. geophys. Res.,* **88,** 4321–4328.

Dupont-Nivet, G., Horton, B.K., Butler, R.F., Wang, J., Zhou, J. & Waanders, G..L., 2004a. Paleogene clockwise tectonic rotation of the Xining-Lanzhou region, northeastern Tibetan Plateau, *J. geophys. Res.,* **109,** B04401, doi:10.1029/2003JB002620.

Dupont-Nivet, G., Butler, R.F., Yin, A. & Chen X., 2003. Paleomagnetism indicates no Neogene vertical axis rotations of the northeastern Tibetan Plateau, *J. geophys. Res.,* **108**(B8), 2386, doi:10.1029/2003JB002399.

Dupont-Nivet, G., Robinson, D., Butler, R.F., Yin, A. & Melosh, H.J., 2004b. Concentration of crustal displacement along a weak Altyn Tagh fault: evidence from paleomagnetism of the northern Tibetan Plateau, *Tectonics,* **23,** TC1020, doi:10.1029/2002TC001397.

Enkin, R.J., 2003. The direction-correction tilt test: an all-purpose tilt/fold test for paleomagnetic studies, *Earth Planet. Sci. Lett.,* **212,** 151–166.

Enkin, R.J., Chen, Y., Courtillot, V., Besse, J., Xing, L., Zhang, Z., Zhuang, Z. & Zhang, J., 1991a. A Cretaceous pole from South China, and the Mesozoic hairpin turn of the Eurasian apparent polar wonder path, *J. geophys. Res.,* **96,** 4007–4027.

Enkin, R.J., Courtillot, V., Xing, L., Zhang, Z., Zhuang, Z. & Zhang, J., 1991b. The stationary Cretaceous paleomagnetic pole of Sichuan (South China Block), *Tectonics,* **10,** 547–559.

Evans, M.A., Lewchuk, M.T. & Elmore R.D., 2003. Strain partitioning of deformation mechanisms in limestones: examining the relationship of strain and anisotropy of magnetic susceptibility (AMS), *J. Struct. Geol.,* **25,** 1525–1549.

Fisher, R.A., 1953. Dispersion on a sphere. *Proc. R. Soc. Lond., A,* **217,** 295–305.

Frost, G.M. *et al*., 1995. Preliminary Early Cretaceous paleomagnetic results from the Gansu Corridor, China, *Earth planet. Sci. Lett.,* **129,** 217–232.

Funahara, S., Nishiwaki, N., Miki, M., Murata, F., Otofuji, Y. & Wang, Y.Z., 1992. Paleomagnetic study of Cretaceous rocks from the Yangtze block, central Yunnan, China: implications for the India-Asia collision, *Earth planet. Sci. Lett.,* **113,** 77–91.

Funahara, S., Nishiwaki, N., Murata, F., Otofuji, Y. & Wang, Y.Z., 1993. Clockwise rotation of the Red River fault inferred from paleomagnetic study of Cretaceous rocks in the Shan-Thai-Malay block of western Yunnan, China, *Earth planet. Sci. Lett.,* **117,** 29–42.

Gan, W., Zhang, P., Shen, Z., Niu, Z., Wang, M., Wan, Y., Zhou, D. & Cheng. J., 2007. Present-day crustal motion within the Tibetan Plateau inferred from GPS measurements, *J. geophys. Res.,* **112,** B08416, doi:10.1029/2005JB004120.

Gareces, M., Pares, J.M. & Cabrera, L., 1996. Further evidence for inclination shallwoing in red beds, *Geophys. Res. Lett.,* **23,** 2065–2068.

Gilder, S.A, Coe. R.S., Wu, H., Kuang, G., Zhao, X., Wu, Q. & Tang, X., 1993. Cretaceous and Tertiary paleomagnetic results form southeast China and their tectonic implications, *Earth Planet. Sci. Lett.,* **117,** 637–652.

Gilder, S.A., Zhao, X.X., Coe, R.S., Meng, Z., Courtillot, V. & Besse, J., 1996. Paleomagnetism and tectonics of the southern Tarim basin, northwestern China, *J. geophys. Res.,* **101,** 22 015–22 031.

Gilder, S.A. *et al.*, 1999. Tectonic evolution of the Tancheng-Lujiang (Tan-Lu) fault via Middle Triassic to Early Cenozoic paleomagnetic data, *J. geophys. Res.,* **104,** 15 365–15 390.

Gilder, S.A., Chen, Y., Cogné, J.-P., Tan, X., Courtillot, V., Sun, D. & Li. Y., 2003. Paleomagnetism of Upper Jurassic to Lower Cretaceous volcanic and sedimentary rocks from the western Tarim Basin and implications for inclination shallowing and absolute dating of the M-0 (ISEA?) chron, *Earth planet. Sci. Lett.,* **206,** 587–600.

Halim, N. *et al*., 1998. New Cretaceous and Early Tertiary paleomagnetic results from Xining-Lanzhou basin, Kunlun and Qiangtang blocks, China: implications on the geodynamic evolution of Asia, *J. geophys. Res.,* **103,** 21 025–21 045.

Hankard, F., Cogne, J-P., Quidelleur, X. & Bayasgalan, A., 2007a. Palaeomagnetism and K-Ar dating of Cretaceous basalts from Mongolia, *Geophys. J. Int.,* **169,** 898–908.

Hankard, F., Cogne, J-P., Kravchinsky, V.A., Carporzen, L., Bayasgalan, A. & Lkhagvadorj, P., 2007b. New Tertiary paleomagnetic poles from Mongolia and Siberia at 40, 30, 20, and 13 Ma: clues on the inclination shallowing problem in central Asia, *J. geophys. Res.,* **112,** B02101, doi:10.1029/2006JB004488, 2007.

Heslop, D., 2007. Are dydrodynamic shape effects important when modeling the formation of depositional remanent manetization? *Geophys. J. Int.,* **171,** 1029–1035.

Horton, B.K., Dupont-Nivet, G., Zhou, J., Waanders, G.L., Butler, R.F. & Wang, J., 2004. Mesozoic-Cenozoic evolution of the Xining-Minhe and Dangchang basins, northeastern Tibetan Plateau: magnetostratigrapic and biostratigraphic results, *J. geophys. Res.,* **109,** B04402, doi:10.1029/2003JB002913.

Houseman, G.A. & England, P.C., 1993. Crustal thickening versus lateral expulsion in the Indian-Asian continental collision, *J. geophys. Res.,* **98,** 12 233–12 249.

Huang, B., Piper, J.D.A., Wang, Y., He, H. & Zhu, R., 2005. Paleomagnetic and geochronological constraints on the post-collisioinal northward convergence of the southwest Tian Shan, NW China, *Tectonophysics,* **409,** 107–124.

Huang, K. & Opdyke, N.D., 1992. Paleomagnetism of Cretaceous to lower Tertiary rocks from Southwestern Sichuan: a revisit, *Earth planet. Sci. Lett.,* **112,** 29–40.

Huang, K. & Opdyke, N.D., 1993. Paleomagnetic results from Cretaceous and Jurassic rocks of South and Southwest Yunnan: evidence for large clockwise rotations in the Indochina and Shan-Thai-Malay terranes, *Earth planet. Sci. Lett.,* **117,** 507–524.

Huang, K., Opdyke, N.D., Li, J.G. & Peng, X.J., 1992. Paleomagnetism of Cretaceous rocks from eastern Qiangtang terrane of Tibet, *J. geophys. Res.,* **97,** 1789–1799.

Hodych, J.P. & Hayatsu, A., 1988. Paleomagnetism and K-Ar isochron dates of Early Jurassic basaltic flows and dikes of Atlantic Canada, *Can. J. Earth Sci.,* **25,** 1972–1989.

Jezek, J. & Gilder, S.A., 2006. Competition of magnetic and hydrohynamic forces on ellipsoidal particles under shear: influence of the Earth's magnetic field on particle alignment in vscous media, *J. geophsy. Res.,* **111,** B12S23, doi:10.1029/2006JB004541.

Kent, D.V., Xu, G., Huang, K., Zhang, W.Y. & Opdyke, N.D., 1986. Paleomagnetism of upper Cretaceous rocks from South China, *Earth planet. Sci. Lett.,* **79,** 179–184.

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Leloup, P.H. *et al*., 1995. The Ailao Shan-Red River shere zone (Yunnan, China), Tertiary transform boundary of Indochina, *Tectonophysics,* **251,** 3–84.

Li, X., 2000. Cretaceous magmatism and lithospheric extension in Southeast China, *J. Asian Earth Sci.,* **18,** 293–305.

Li, Y., Zhang, Z., McWilliams, M., Sharps, R., Zhai, Y., Li. Y., Li, Q. & Cox, A., 1988. Mesozoic Paleomagnetic results of the Tarim craton: Tertiary relative motion between China and Siberia? *Geophys. Res. Lett.,* **15,** 217–220.

Li, Y., Ali, J.R., Chan, L.S.& Lee, C.M., 2005. New and revised set of Cretaceous paleomagnetic poles from Hong Kong: implications for the development of southeast China. *J. Asian Earth Sci.,* **24,** 481–493.

Li, Z.X., Metcalfe, I., & Wang, X., 1995. Vertical-axis block rotations in southwestern China since the Cretaceous: new paleomagnetic results from Hainan Island, *Geophys. Res. Lett.,* **22,** 3071–3074.

Lin, J. & Watts, D.R., 1988. Palaeomagnetic results from the Tibetan Plateau, *Phil. Trans. R. Soc. Lond., A,* **327,** 239–262.

Liu, Y. & Morinaga, H., 1999. Cretaceous palaeomagnetic results from Hainan Island in south China supporting the extrusion model of Southeast Asia, *Tectonophys.*, **301,** 133–144.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett.,* **17,** 159–162.

Lu, H. & Xiong, S., 2009. Magnetostratigraphy of the Dahonggou section, northern Qaidam Basin and its bearing on Cenozoic tectonic evolution of the Qilian Shan and Altyn Tagh Fault, *Earth planet. Sci. Lett.,* **288,** 539–550.

Macmillan, S. & Maus, S., 2005. International geomagnetic reference field-the tenth generation, *Earth Planets Space,* **57,** 1135–1140.

McFadden, P.L., 1990. A new fold test for palaeomagnetic studies, *Geophys. J. Int.* **103,** 163–169.

McFadden, P.L. & McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism, *Geophys. J. Int.* **103,** 725–729.

Meng, Q-R. & Zhang, G-W., 2000. Geologic framework and tectonic evolution of the Qinling orogen, central China, *Tectonophysics,* **323,** 183–196.

Meng, Q-R., Wang, E. & Hu, J-M., 2005. Mesozoic sedimentary evolution of the northwest Sichuan basin: implication for continued clockwise rotation of the South China block, *Geol. Soc. Am. Bull.,* **117,** 396–410.

Mitra, R. & Tauxe, L., 2009. Full vector model for magnetization in sediments, *Earth planet. Sci. Lett.,* **286,** 535–545.

Morinaga, H. & Liu, Y., 2004. Cretaceous paleomagnetism of the eastern South China Block: establishment of the stable body of SCB. *Earth planet. Sci. Lett.,* **222,** 971–988.

Narumoto, K., Yang, Z., Takemoto, K., Zaman, H., Morinaga, H. & Otofuji, Y., 2006. Anomalously shallow inclination in middle-northern part of the South China block: palaeomagnetic study of Late Cretaceous red beds from Yichang area. *Geophys. J. Int.,* **164,** 290–300.

National Geophysical Data Center, 2001. ETOPO2 Global 2′ Elevations, *National Oceanic and Atmospheric Administration,* U.S. Department of Commerce, Boulder, CO (CD-ROM).

Otofuji, Y., Funahara, S., Matsuo, J., Murata, F., Nishiyama, T., Zheng, X. & Yaskawa, K., 1989. Paleomagnetic study of western Tibet; deformation of a narrow zone along the Indus Zhangbo suture between India and Asia, *Earth planet. Sci. Lett.,* **92,** 307–316.

Otofuji, Y., Inoue, Y., Funahara, S., Murata, F. & Zheng, X., 1990. Palaeomagnetic study of eastern Tibet—deformation of the Three Rivers region, *Geophys. J. Int.,* **103,** 85–94.

Otofuji, Y., Kadoi, J., Funahara, S., Murata, F. & Zheng, X, 1991. Paleomagnetic study of the Eocene Quxu pluton of the Gandese Belt: crustal deformation along the Indus-Zangbo suture zone in southern Tibet, *Earth planet. Sci. Lett.,* **107,** 369–379.

Otofuji, Y., Liu, Y., Yokoyama, M., Tamai, M. & Yin, J., 1998. Tectonic deformation of the southwestern part of the Yangtze craton inferred from paleomagnetism, *Earth planet. Sci. Lett.,* **156,** 47–60.

Otofuji, Y. *et al*., 2007. Spatial gap between Lhasa and Qiangtang blocks inferred from Middle Jurassic to Cretaceous paleomagnetic data, *Earth planet. Sci. Lett.,* **262,** 581–593.

Otofuji, Y., Yokoyama M., Kitada K. & Zaman H., 2010. Paleomagnetic versus GPS determined tectonic rotation around eastern Himalayan syntaxis in East Asia, *J. Asian Earth Sci.,* **37,** 438–451.

Payen, M.A. & Verosub, K.L., 1982. The acquisition of post-depositional detrital remanent magnetization in a varity of natural sediments, *Geophys. J. R. astr. Soc.,* **68,** 625–642.

Pozzi, J.P., Westphal, M., Girardeau, J. Besse, J., Zhou, Y.X., Chen, X.Y. & Xing, L.S., 1984. Paleomagnetism of the Xigaze ophiolite and Flysh (Yarlung Zangbo suture zone, south Tibet): latitude and direction of spreading, *Earth planet. Sci. Lett.,* **70,** 383–394.

Richter, B. & Fuller, M., 1996. Palaeomagnetism of the Sibumasu and Indochina blocks: implications for the extrusion tectonic models, *Geol. Soc. London Spec. Pub.*, **106**, 203–224.

Ritts, B.D. & Biffi, U., 2001. Mesozoic northeast Qaidam basin: response to contractional reactivation of the Qaidam Shan, and implications for the extent of Mesozoic intracontinental deformation in central Asia, *Geol. Soc. Am. Mem.*, **194**, 293–316.

Roger, F. *et al*., 2003. Geochronological and geochemical constraints on Mesozoic suturing in east central Tibet, *Tectonics,* **22,** 1037 doi:10.1029/2002TC001466.

Royden, L.H., Burchfiel, B.C. & Van Der Hilst, R.D., 2008. The geological evolution of the Tibetan Plateau. *Science,* **321,** 1054–1058.

Rumelhart, P.E., Yin, A., Cowgill, E., Butler, R., Zhang, Q. & Wang, X-F., 1999. Cenozoic vertical-axis rotation of the Altyn Tagh fault system, *Geology,* **27,** 819–822.

Sato, K., Liu, Y.Y., Zhu, Z.C., Yang, Z.Y. & Otofuji, Y., 1999. Paleomagnetic study of middle Cretaceous rocks from Yunlong, western Yunnan, China: evidence of southward displacement of Indochina, *Earth planet. Sci. Lett.,* **165,** 1–15.

Sato, K., Liu, Y., Wang, Y., Yokoyama, M., Yoshioka, S., Yang, Z. & Otofuji, Y., 2007. Paleomagnetic study of Cretaceous rocks from Pu'er, western Yunnan, China: evidence of internal deformation of the Indochina block, *Earth planet. Sci. Lett.,* **258,** 1–15.

Shu, L.S., Zhou, X.M., Deng, P., Wang, B., Jiang, S.Y., Yu, J.H. & Zhao, X.X., 2009. Mesozoic tectonic evolution of the Southeast China Block: new insights from basin analysis, *J. Asian Earth Sci.,* **34,** 376–391.

Smith, W.H.F. & Wessel, P., 1990. Gridding with continuous curvature splines in tension, *Geophysics* **55,** 293–305.

Su, L., Yang, Z. Sun, Z., Yang, T., Zaman, H., Takemoto, K. & Y. Otofuji, Y., 2005. Regional deformational features of the South China Block inferred from Middle Triassic palaeomagnetic data, *Geophys. J. Int.,* **162,** 339–356.

Stamatakos, J., Lessard, A.M., Van Der Pluijm, B.A. & Van Der Voo, R., 1995. Paleomagnetism and magnetic fabrics from the Springdale and Wigwam Redbeds of Newfoundland and their implications for the Silurian paleolatitute controversy, *Earth planet. Sci. Lett.,* **132,** 141–155.

Sun, Z., Yang, Z., Pei, J., Yang, T. & Wang, X., 2006a. New Early Cretaceous paleomagnetic data from volcanic and red beds of the eastern Qaidam block and its implications for tectonics of Central Asia, *Earth planet. Sci. Lett.,* **243,** 268–281.

Sun, Z., Yang, Z., Yang, T., Pei, J. & Yu, Q., 2006b. New Late Cretaceous and Paleogene paleomagnetic results from South China and their geodynamic implications, *J. geophys. Res.,* **B111**(B3), B03101, doi:10.1029/2004JB0034555.

Takemoto, K., Halim, N., Otofuji, Y., Tran, V.T., Le V.D. & Hada, S., 2005. New paleomagnetic constraints on the extrusion of Indochina: Late Cretaceous results from the Song Da terrane, northern Vietnam, *Earth planet. Sci. Lett.,* **229,** 273–285.

Takemoto, K. *et al*., 2009. Tectonic deformation of the Indochina Peninsula recorded in the Mesozoic palaeomagnetic results, *Geophys. J. Int.,* **179**, 97–111.

Tamai, M., Liu, Y.Y., Lu, L.Z., Yokoyama, M., Halim, N., Zaman, H. & Otofuji, Y., 2004. Palaeomagnetic evidence for southward displacement of the Chuan Dian Fragment of the Yangtze block. *Geophys. J. Int.,* **158,** 297–309.

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025

Tan, X., Kodama, K. & Fang, D., 2002. Laboratory depositional and compaction-caused inclination errors carried by haematite and their implications in identifying inclination error of natural remanence in red beds, *Geophys. J. Int.,* **151,** 475–486.

Tan, X., Kodama, K., Chen, H., Fang, D., Sun, D. & Li, Y., 2003. Paleomagnetism and magnetic anisotropy of Cretaceous red beds from the Tarim basin, northwest China: evidence for a rock magnetic cause of anomalously shallow paleomagnetic inclinations from central Asia, *J. geophys. Res.,* **108**(B2), 2107, doi:10.1029/2001JB001608.

Tan, X., Kodama K.P., Gilder S. & Courtillot V., 2007. Rock magnetic evidence for inclination shallowing in the Passaic Formation red beds from the Newark basin and a systematic bias of the Late Triassic apparent polar wander path for North America, *Earth planet. Sci. Lett.,* **254,** 345–357.

Tan, X., Gilder, S., Kodama, K., Jian, W., Han, Y., Zhang, H.M., Xu, H. & Zhou, D., 2010. New paleomagnetic results from the Lhasa block: revised estimation of latitudinal shortening across Tibet and implications for dating the India-Asia collision, *Earth planet. Sci. Lett.,* **293,** 396–404.

Tanaka, K. *et al*., 2008. Tectonic deformation around the eastern Himalayan syntaxis: constraints from the Cretaceous palaeomagnetic data of the Shan-Thai Block, *Geophys. J. Int.,* **175,** 713–728.

Tapponnier, P., Peltzer, G., Le Dain, A.Y., Armijo, R. & Cobbold, P., 1982. Propagating extrusion tectonics in Asia: new insights from simple experiments with plasticine. *Geology* **10,** 611–616.

Tapponnier, P., Xu Z., Roger, F., Meyer, B., Arnaud, N., Wittlinger, G. & Yang J., 2001. Oblique Stepwise Rise and Growth of the Tibet Plateau, *Science,* **294,** 1671–1677.

Tarling, D.H. & Hrouda, F., 1993. *The Magnetic Anisotropy of Rocks,* Chapman & Hall, London, 217.

Tauxe, L., 2005. Inclination flattening and the geocentric axial dipole hypothsis, *Earth planet. Sci. Lett.,* **233,** 247–261.

Tauxe, L. & Kent, D.V., 1984. Properties of a detrital remanence carried by haematite from study of modern river deposits and laboratory redeposition experiments, *Geophys. J. R. astr. Soc.,* **76,** 543–561.

Tauxe, L. & Kent, D.V., 2004. A simplified statistical model for the geomagnetic field and the detection of shallow bias in paleomagnetic inclinations: was the ancient magnetic field dipolar?, *Geophys. Monogr.*, **145,** 101–115

Tauxe, L., Kodama, K.P. & Kent, D.V., 2008. Testing corrections for paleomagnetic inclination eror in sedimentary rocks: a comparative approach, *Phys. Earth planet. Int.,* **169,** 152–165.

Thompson, R., 1986. Modelling magnetization data using SIMPLEX, *Phys. Earth planet. Inter.,* **42,** 113–127.

Tsuneki, Y., Morinaga, H. & Liu, Y., 2009. New palaeomagnetic data supporting the extent of the stable body of the South China Block since the Cretaceous and some implications on magnetization acquisition of red beds, *Geophys. J. Int.,* **178**, 1327–1336.

Tucker, P., 1979. Selective post-depositional realignment in a synthetic sediment, *Phys. Earth planet. Int.,* **20,** 11–14.

Van Der Voo, R. & Torsvik, T.H., 2001. Evidence for late Paleozoic and Mesozoic non-dipole fields provides an explanation for the Pangea reconstruction problems, *Earth planet. Sci. Lett.,* **187,** 71–81.

Wang, B. & Yang, Z., 2007. Late Cretaceous paleomagnetic results from southeastern China, and the geological implication, *Earth planet. Sci. Lett.,* **258,** 315–333.

Wessel, P. & Smith, W.H.F., 1991. Free software helps map and display data, *EOS, Trans. Am. geophys. Un.,* **72,** 441–446.

Westphal, M., 1993. Did a large departure from the geocentric axial dipole hypothesis occur during the Eocene? Evidence from the magnetic polar wander path of Eurasia, *Earth planet. Sci. Lett.,* **117,** 15–28.

Westphal, M., Bazhenov, M.L., Lauer, J.P., Pechersky, D.M. & Shibuet, J.C., 1986. Paleomagnetic implications on the evolution of the Tethys belt from the Atlantic Ocean to the Permirs since the Triassic, *Tectonophysics,* **123,** 37–82.

Yang, Z.Y. & Besse, J., 1993. Paleomagnetic study of Permian and Mesozoic sedimentary rocks from Northern Thailand supports the extrusion model for Indochina, *Earth planet. Sci. Lett.,* **117,** 525–552.

Yang, Z.Y., Yin, J.Y., Sun, Z..M., Otofuji, Y. & Sato, K., 2001. Discrepant Cretaceous paleomagnetic poles between Eastern China and Indochina: a consequence of the extrusion of Indochina, *Tectonophysics,* **334,** 101–113.

Yin, A. & Harrison, T.M., 2000. Geologic evolution of the Himalayan-Tibetan Orogen, *Annu. Rev. Earth planet. Sci.,* **28,** 211–280.

Yin, A., Manning, C.E., Lovera, O., Menold, C.A., Chen, X. & Gehrels, G.E., 2007. Early Paleozoic Tectonic and Thermomechanical Evolution of Ultrahigh-Pressure (UHP) Metamorphic Rocks in the Northern Tibetan Plateau, Northwest China, *Int. Geol. Rev.,* **49,** 681–716.

Yokoyama, M., Liu, Y., Otofuji, Y. & Yang, Z., 1999. New Late Jurassic paleomagnetic data from the northern Sichuan basin: implications for the deformation of the Yangtze craton, China, *Geophys. J. Int.,* **139,** 795–805.

Yokoyama, M., Liu, Y., Halim, N. & Otofuji, Y., 2001. Paleomagnetic study of Upper Jurassic rocks from the Sichuan basin: tectonic aspects for the collision between the Yangtze block and the North China Block, *Earth planet. Sci. Lett.*, **193,** 273–285.

Yoshioka, S., Liu, Y.Y., Sato, K., Inokuchi, H., Su, L., Zaman, H. & Otofuji, Y., 2003. Paleomagnetic evidence for post-Cretaceous internal deformation of the Chuan Dian Fragment in the Yangtze block: a consequence of indentation of India into Asia, *Tectonophysics,* **376,** 61–74.

Zhu, Z., Morinaga, H., Gui, R., Xu, S. & Liu, Y., 2006. Paleomagnetic constraints on the extent of the stable body of the South China block since the Cretaceous: new data from the Yuanma Basin, China, *Earth planet. Sci. Lett.,* **248,** 533–544.

Zijderveld, J.D.A., 1967. A. C. demagnetization of rocks: analysis of results, in *Methods in Palaeomagnetism,* pp. 254–286, eds Collinson, D.W., Creer, K.M. & Runcorn, S.K., Elsevier, Amsterdam.

Downloaded from https://academic.oup.com/gji/article/186/2/471/586730 by guest on 19 May 2025